



AKTU



B. Tech II-Year

Digital Electronics

Unit-4 in One Shot

Pdf Notes + AKTU PYQs + Imp. Questions



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AKTU B.Tech II-Year

Subject : DIGITAL ELECTRONICS



Unit-4 : Syllabus

Synchronous & Asynchronous Sequential Circuits: Analysis of clocked sequential circuits with state machine designing, State reduction and assignments, Design procedure. Analysis procedure of Asynchronous sequential circuits, circuit with latches, Design procedure, Reduction of state and flow table, Race-free state assignment, Hazards.

Types of Sequential Circuits

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Synchronous

Models

A clocked flipflop acts as memory element

Controlled by clock i.e these are active only when clock is present

Slower due to Clock

Easy to Design

High Cost

Asynchronous

An unclocked flipflop or time delay elements acts as memory element

Latch (o/p changes with change in input, clock independent)

Not controlled by clock i.e these are active as soon as input changes.

Faster as clock is not present

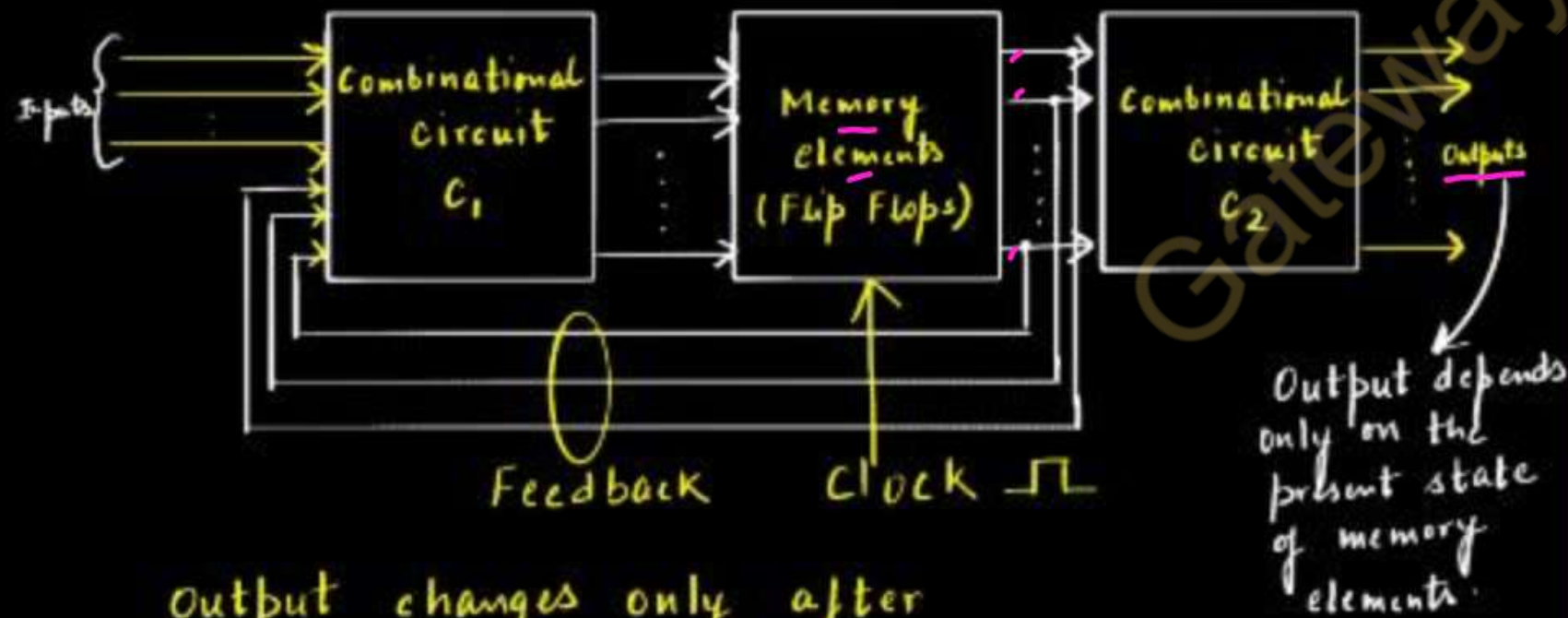
Difficult to design

Low Cost

Moore Circuit

Its output depends only on the **present states** of Flipflops.

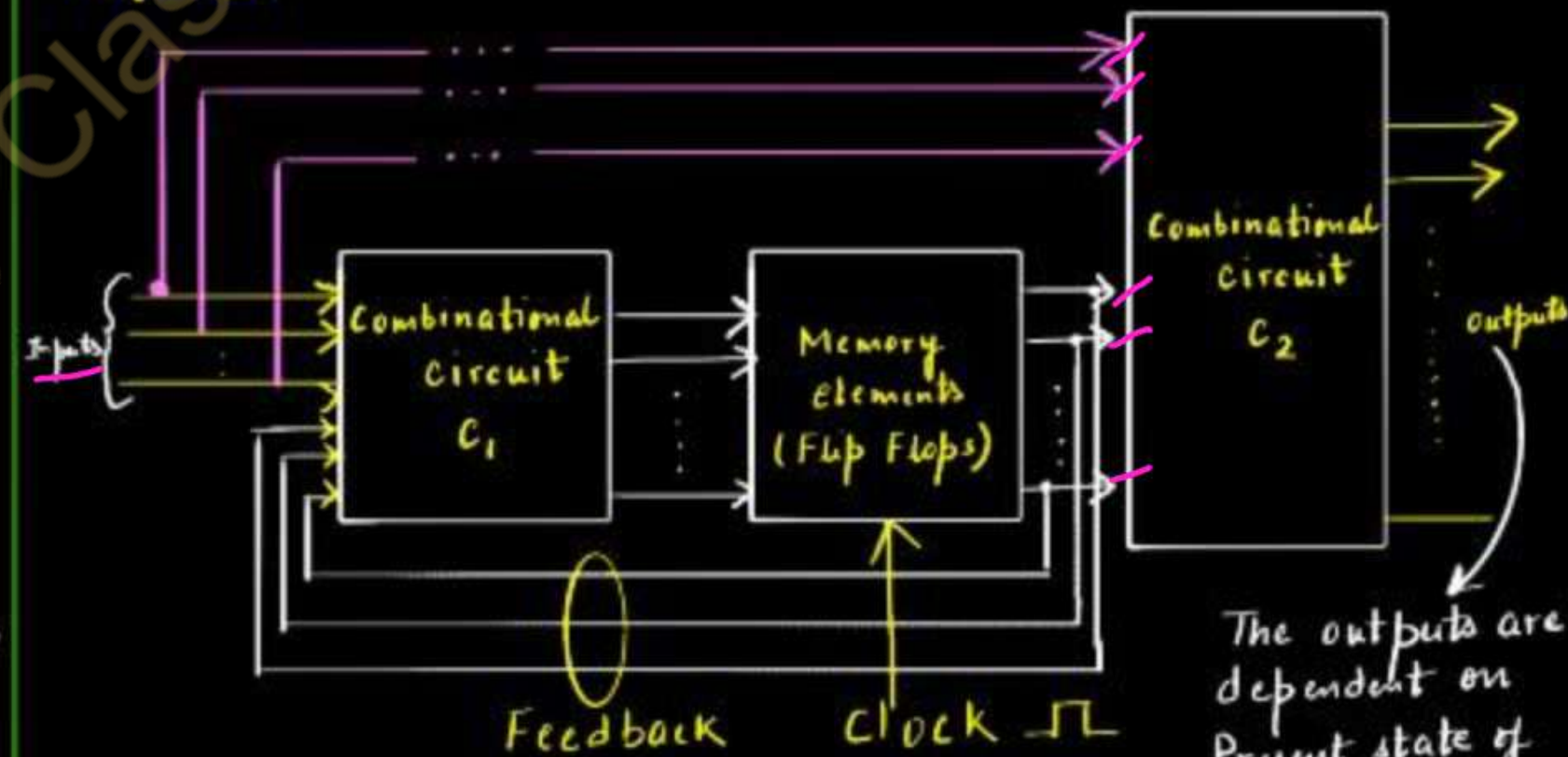
The output does not depend on the external inputs.



output changes only after the active clock edge.

Mealy Circuit

Its output depends only on the **present states** of Flipflops as well as the **external inputs**.



Output can change in between the clock edges if the external inputs change.

Analysis of Clocked (Synchronous) Sequential Circuits

The analysis of the given clocked sequential circuit includes

Writing the State Table

And

Drawing the State Diagram.

State Table

The relation between Inputs and Outputs in a

(i) Combinational Circuit is explained by Truth Table.

(ii) Sequential Circuit is explained by State Table

which includes

Present State, *PS*

Inputs, *X*

Next State, *NS*

Output, *Y*

PS Input NS o/p
00 → 0 00 0
00 → 1 10 1

STATE TABLE

Present State		Next State				Output	
		X=0		X=1		X=0	X=1
<u>A</u>	<u>B</u>	<u>A</u>	<u>B</u>	<u>A</u>	<u>B</u>	<u>Y</u>	<u>Y</u>
<u>0</u>	<u>0</u>	0	0	1	0	0	1
<u>0</u>	<u>1</u>	1	1	0	0	0	0
<u>1</u>	<u>0</u>	1	0	0	1	1	0
<u>1</u>	<u>1</u>	0	0	1	0	1	0

Handwritten annotations on the State Table:

- Arrows from Present State (0,0) to Next State (0,0) and (1,0).
- Arrows from Present State (0,1) to Next State (1,1) and (0,0).
- Arrows from Present State (1,0) to Next State (1,0) and (0,1).
- Arrows from Present State (1,1) to Next State (0,0) and (1,0).
- Next State (0,0) is labeled with a circled 1.
- Next State (1,0) is labeled with a circled 3.
- Output (0,0) is labeled with a circled 2.
- Output (1,0) is labeled with a circled 4.

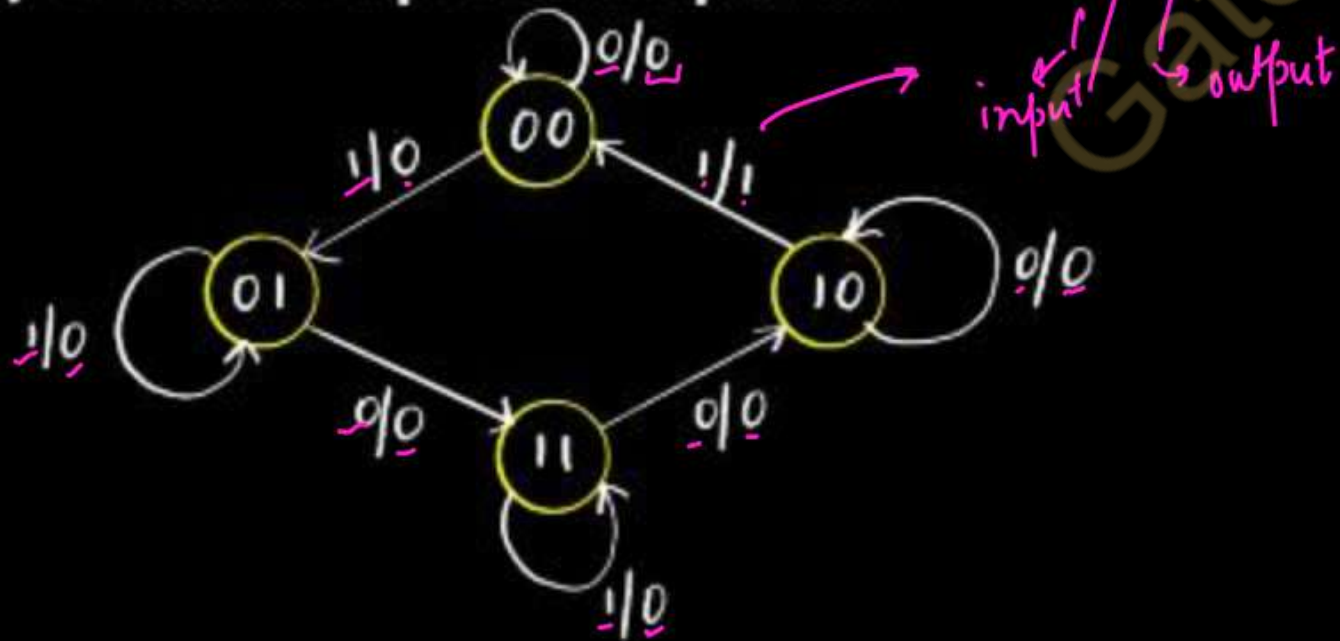
State Diagram

The information available in the state table is represented graphically using the State Diagram.

The **circles** represents the present state. PS

The **arrows** between the circles define the state transition
eg. 00 to 01 or 01 to 11.

The **lines** joining the circles are labeled with a pair of binary numbers with a "/" in between.
Eg. 1/0 means input 1 output 0.



State Equation

It is an algebraic equation.

The Left side represents the next state of the flipflops.

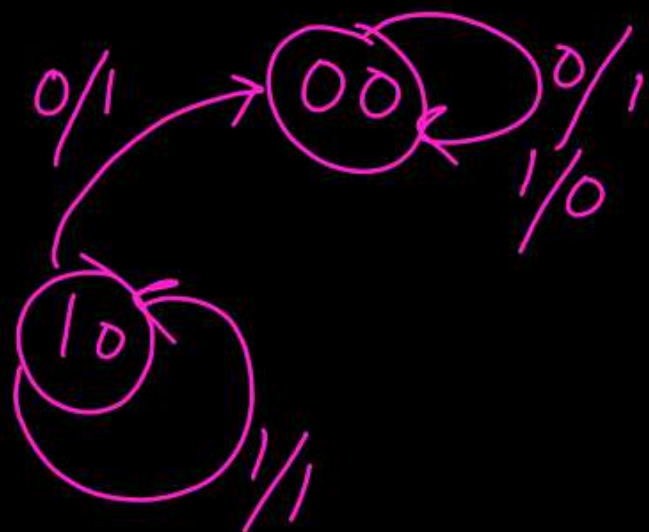
The right hand side specifies the present state conditions

$$Q_{A(n+1)} = \overline{Q}_A Q_B + Q_A \overline{Q}_B + Q_A \overline{Q}_B x \dots$$

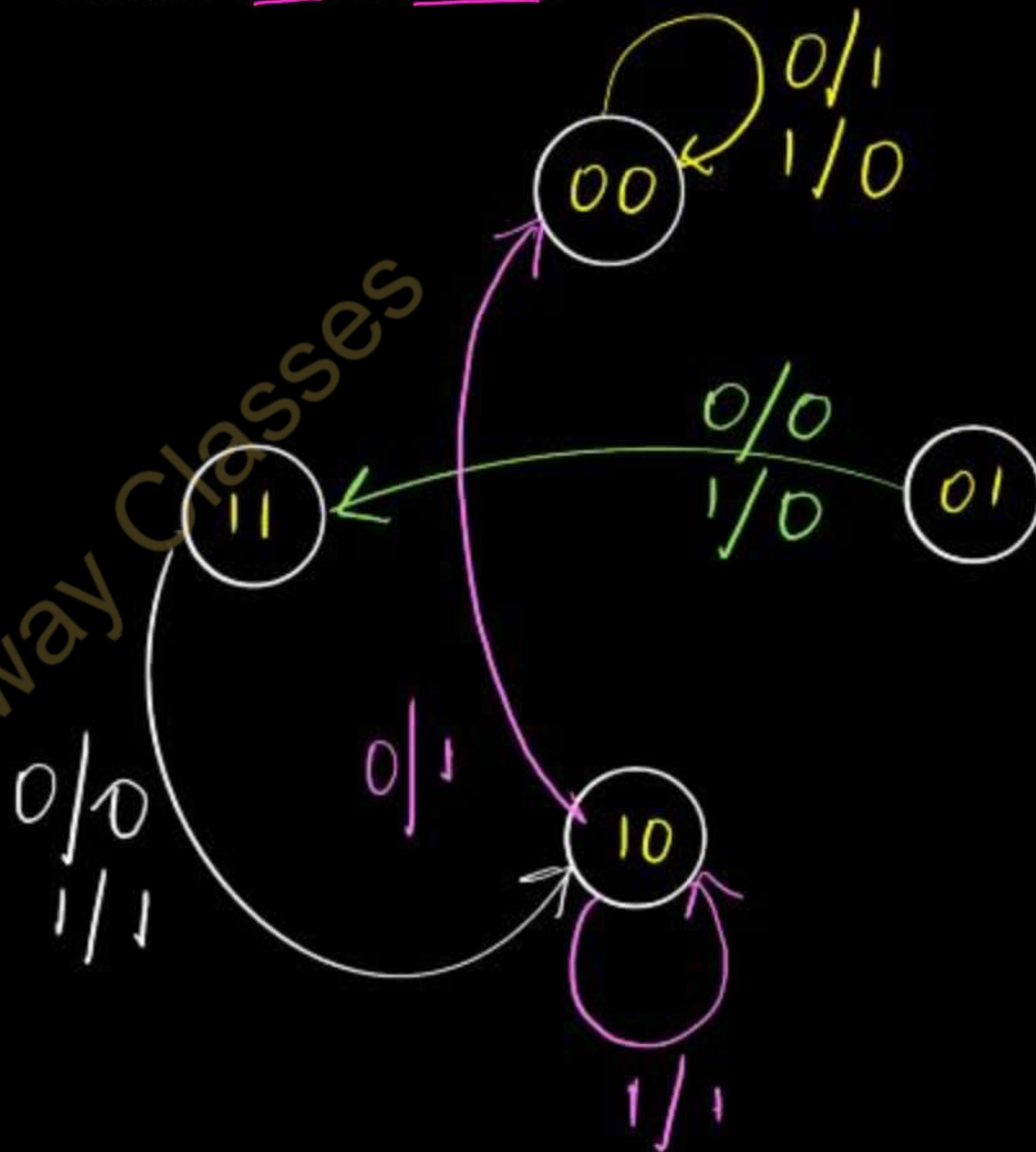
Next state NS

Combinations of present state and inputs

Present State		Next State				Output	
		<u>X=0</u>		<u>X=1</u>		<u>X=0</u>	<u>X=1</u>
A	B	A	B	A	B	Y	Y
<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>1</u>	0
0	1	1	1	1	1	0	0
<u>1</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>1</u>	1
1	1	1	0	1	0	0	1



Step 3 State Diagram



State Reduction

In the design of synchronous sequential circuits the first step is to draw the State Diagram and then write the State Table.

In the state table/state diagram some of the states may be **Equivalent** (same inputs, same next state & same output).

If two states are Equivalent , then one of them is considered as **Redundant State**, and can be **eliminated** in order to reduce the number of Flipflop and Gates used in designing of Sequential Circuits. } imp.

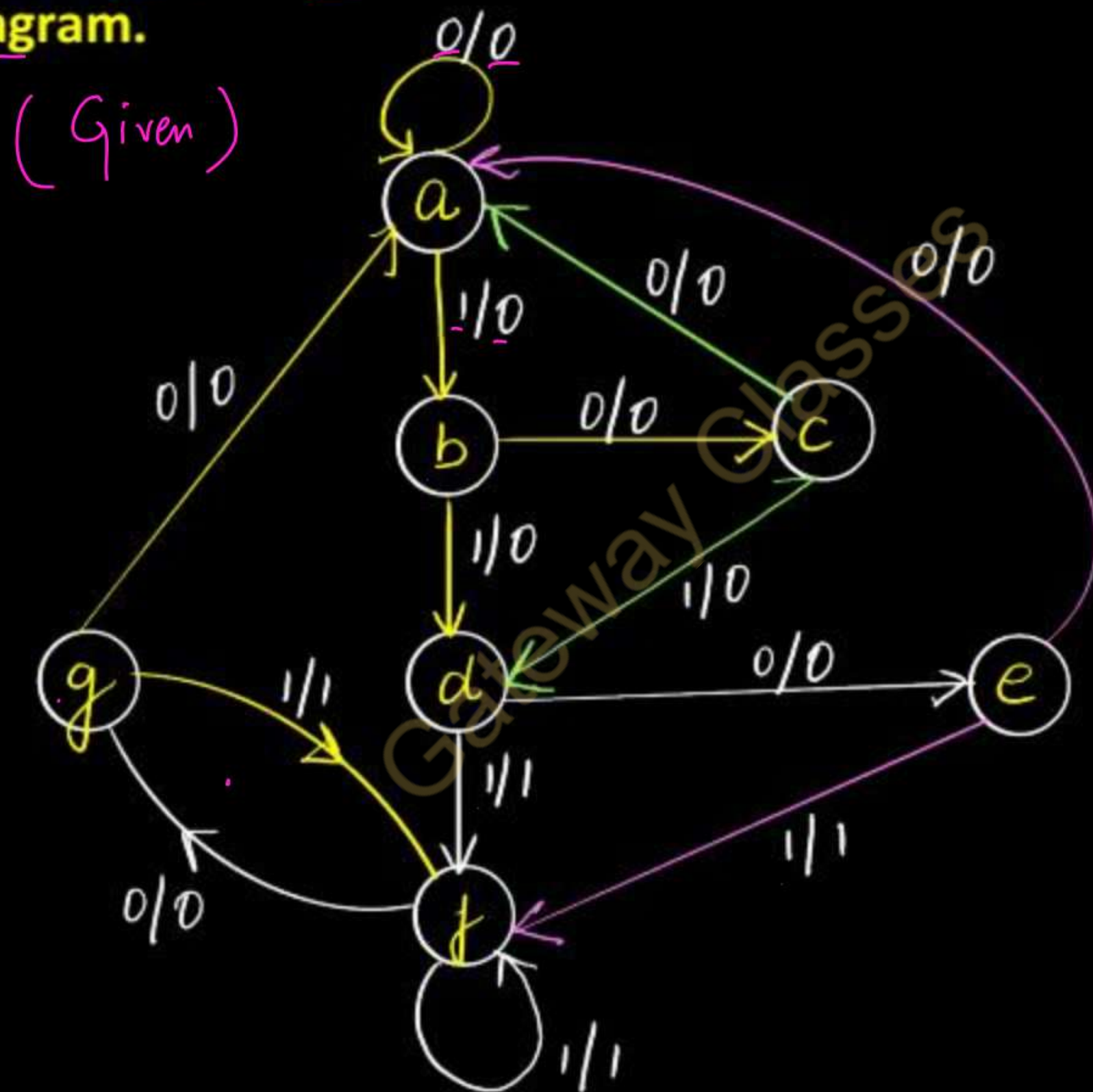
This process of State Elimination of Redundant States is called **State Reduction**.

State Reduction

Steps :-

1. Derive the **State Table** from the given **State Diagram**.
2. Find out the **Equivalent states**.
3. **Eliminate** out of the every pair of equivalent states and write a new State Table.
4. In the new State Table, check for Equivalent states and continue till all the redundant states are eliminated.
5. Draw the **State Diagram with reduced number of states**.

Que-2) Reduce the number of states from the following state diagram . Also draw the reduced state diagram.



It has
7 states

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
→ <u>e</u>	<u>a</u>	<u>f</u>	<u>0</u>	<u>1</u>
f	g	f	0	1
→ <u>g</u>	<u>a</u>	<u>f</u>	<u>0</u>	<u>1</u>

→ states e and g are Equivalent states.

Eliminate row of g and Replace g by e

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g e	f	0	1

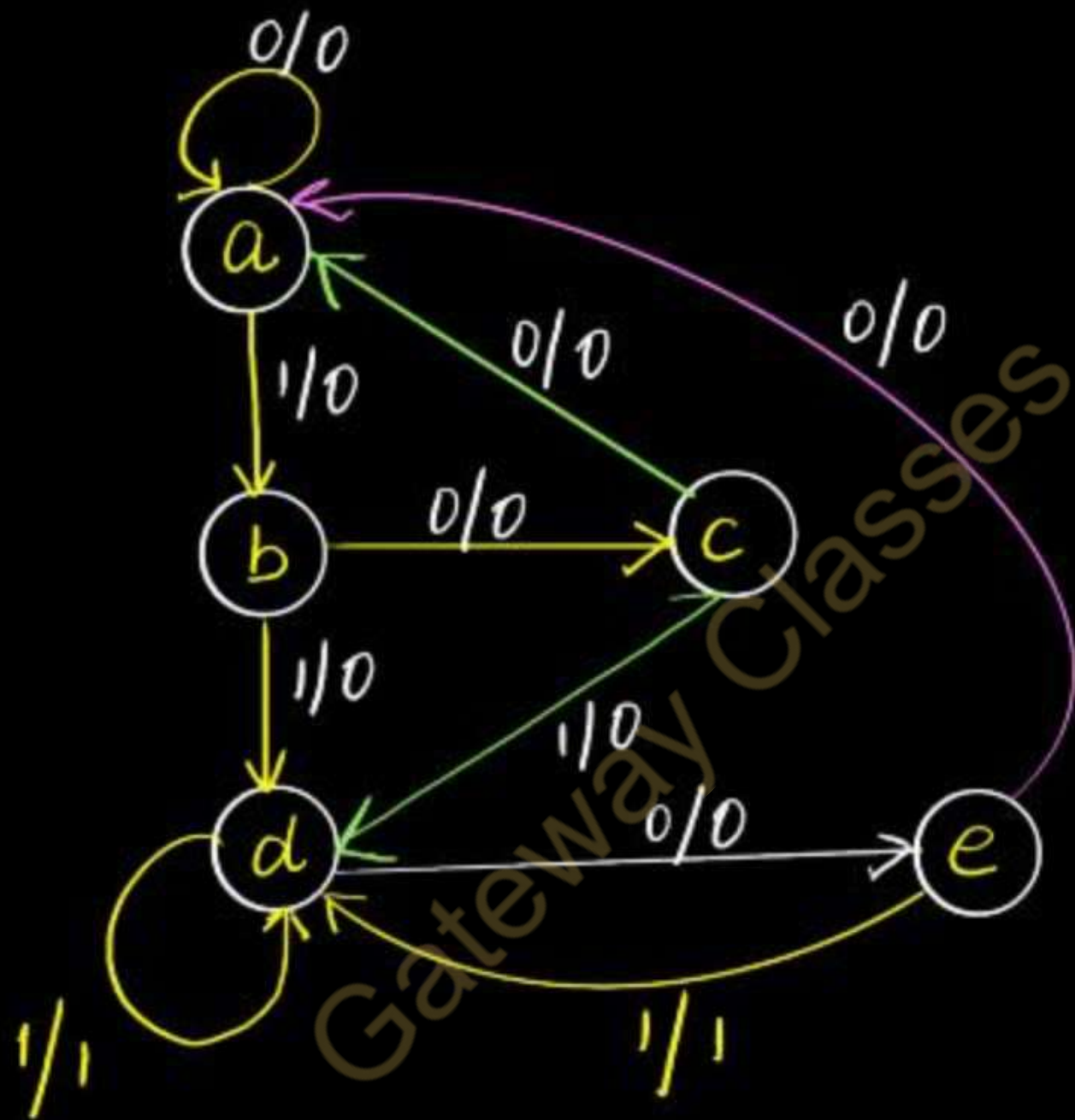
Eliminate row of f and
Replace f by d.

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g e	f	0	1

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
<u>d</u>	<u>e</u>	<u>f</u>	<u>0</u>	<u>1</u>
e	a	f	0	1
<u>f</u>	<u>e</u>	<u>f</u>	<u>0</u>	<u>1</u>

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
<u>a</u>	<u>a</u>	b	<u>0</u>	0
b	c	d	0	0
c	a	d	0	0
d	e	f d	0	1
e	a	f d	0	1

states d and f
are Equivalent
states



(Reduced state Diagram)
↳ has 5 states

State Assignment

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It is the technique in which we **assign binary values to states** in order to reduce the number of gates used and hence **to reduce the cost of the circuit**.

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	d	0	1
e	a	d	0	1

Let

a 001

b 010

c 011

d 100

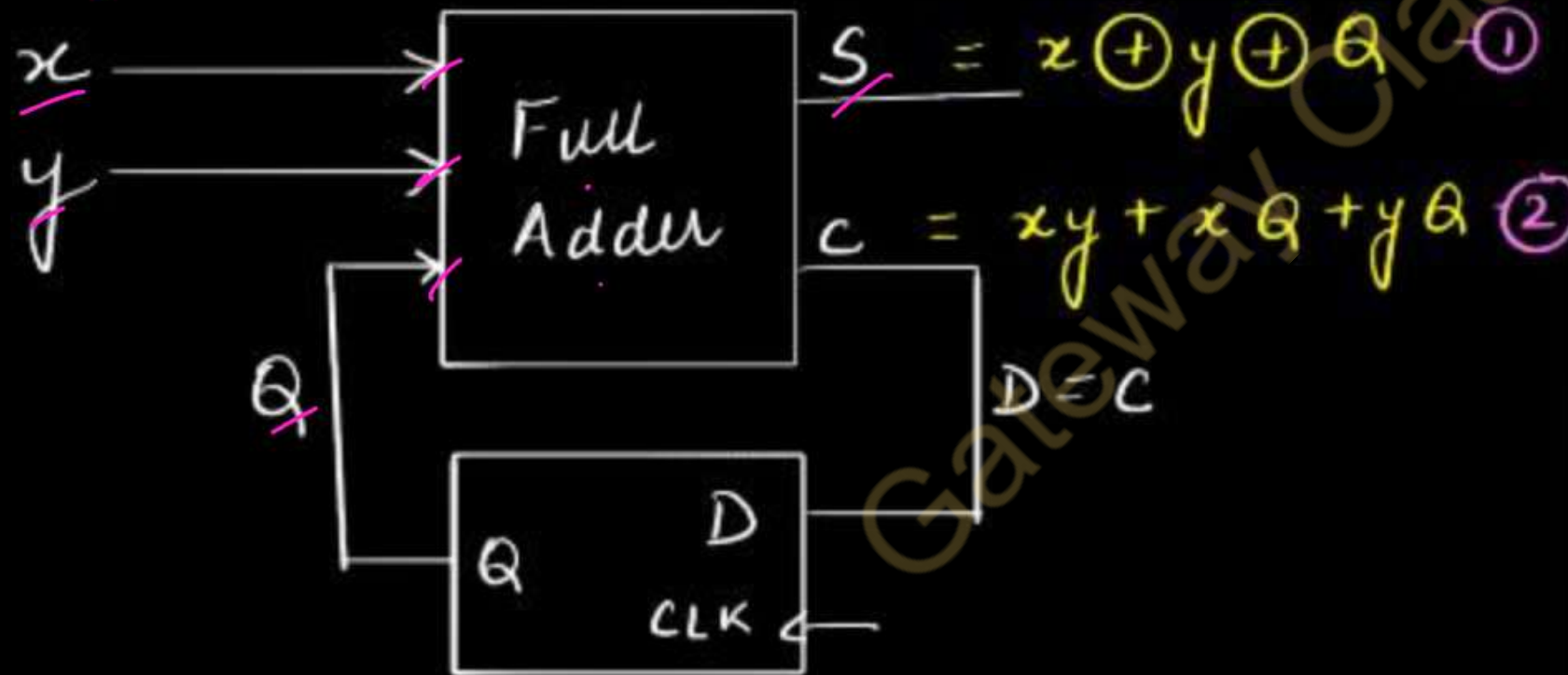
e 101

Present State	Next State		Output	
	X=0	X=1	X=0	X=1
001 a	001	010	0	0
010 b	011	100	0	0
011 c	001	100	0	0
100 d	101	100	0	1
101 e	001	100	0	1

Que 3)

A sequential circuit has one flip-flop Q, two inputs x and y, and one output S. It consists of a full-adder circuit connected to a D flip-flop, as shown in given figure. Derive the state table and state diagram of the sequential circuit. What does this circuit represent?

(Given)



It is a Serial Adder

State Table

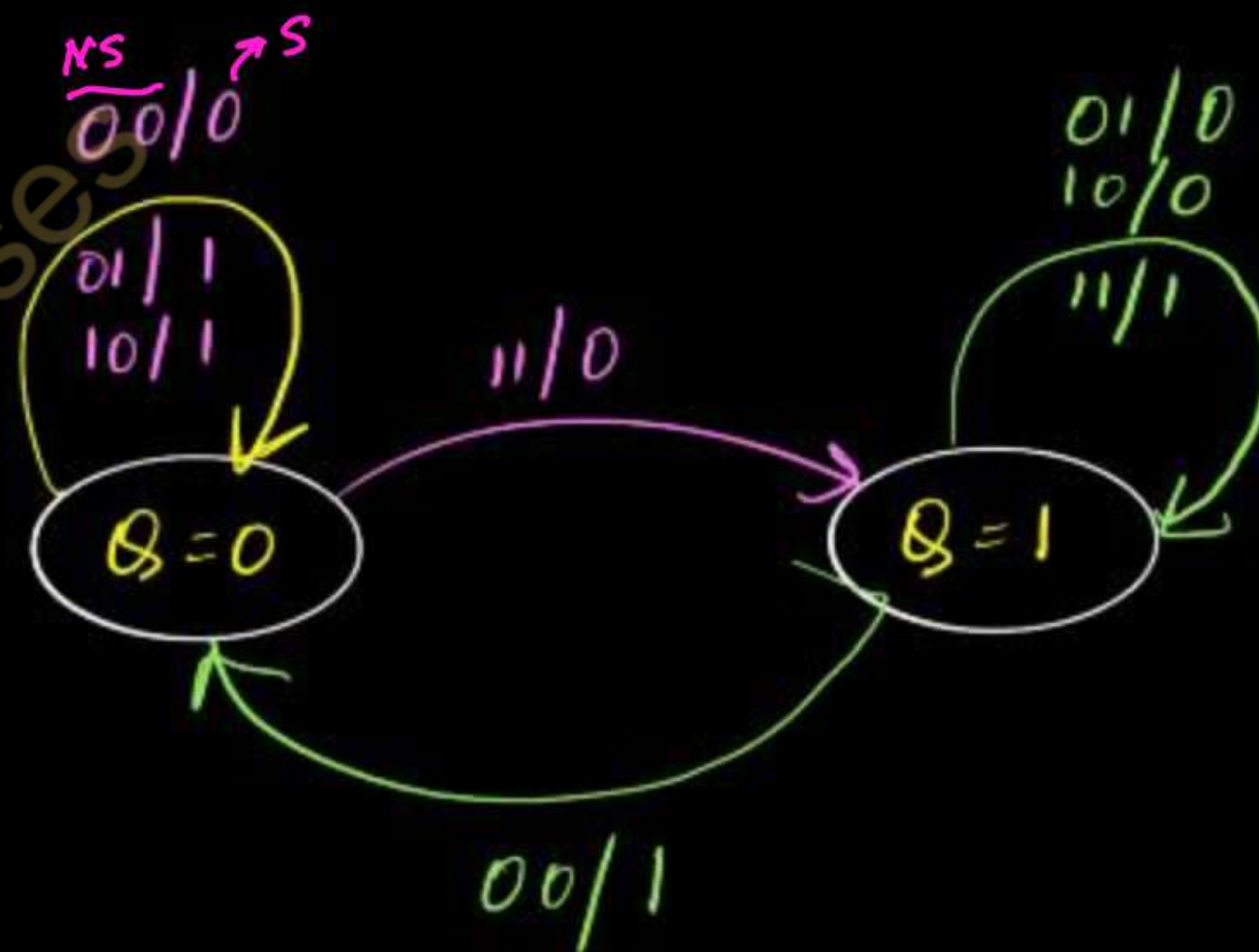
Present State	Inputs		Next State	Output
Q	x	y	D ⁽²⁾	S ⁽¹⁾
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

State Table

Present State	Inputs		Next State	Output
<u>Q</u>	<u>x</u>	<u>y</u>	<u>D</u>	<u>S</u>
Q=0	0	0	0	0
	0	1	0	1
	1	0	0	1
	1	1	1	0
Q=1	0	0	0	1
	0	1	1	0
	1	0	1	0
	1	1	1	1

State Diagram

Two states Q=0 state 0
 Q=1 state 1



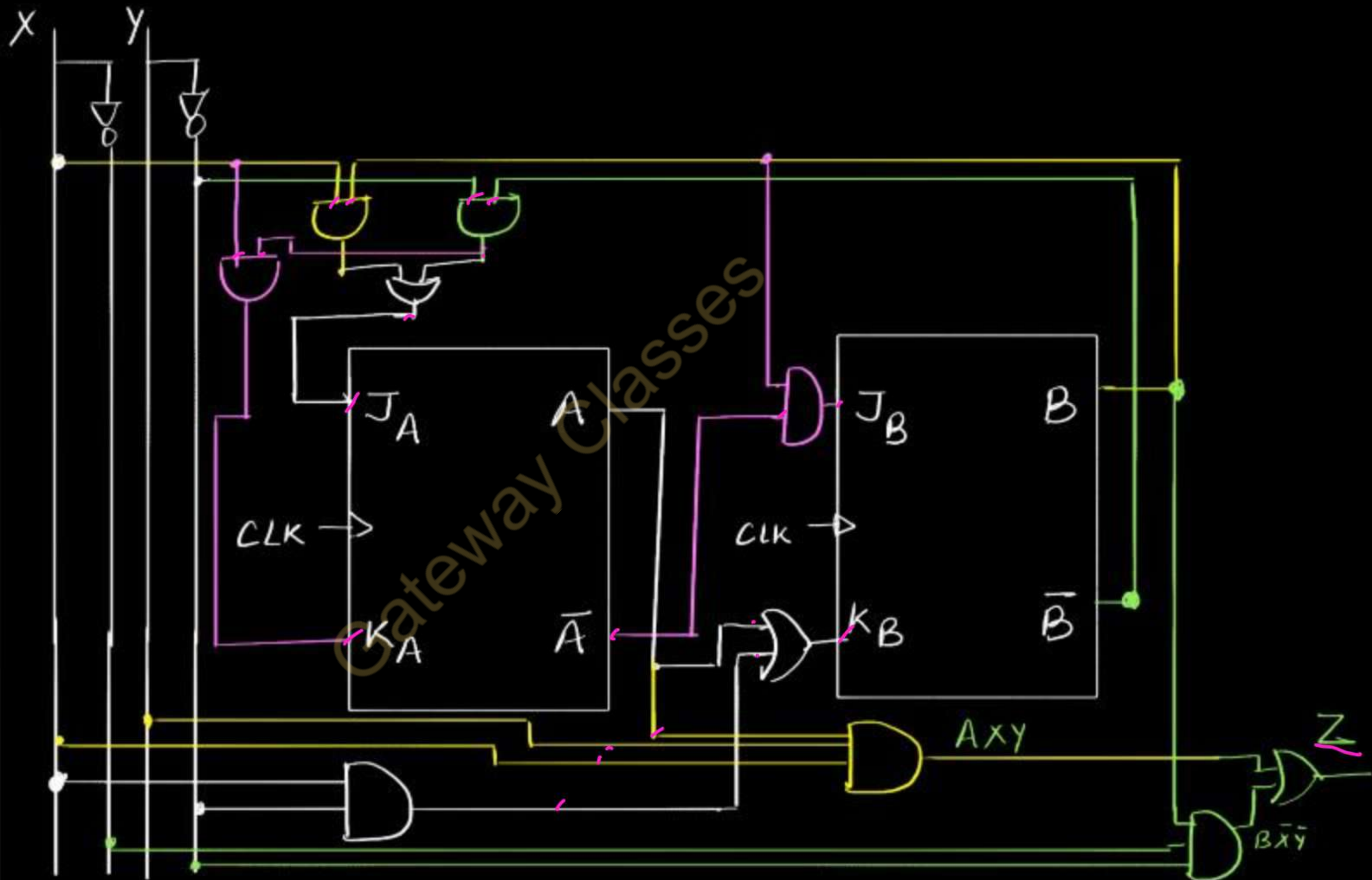
Que 4) A sequential circuit has two J K flip flops A & B, two inputs X & Y, and one output Z. The equations defining this system are as following:

$$\begin{cases} J_A = BX + B'Y' \\ K_A = B'XY' \\ J_B = A'X \\ K_B = A + XY' \\ Z = AXY + BX'Y' \end{cases}$$

Design the circuit.

Gateway Classes

$$\begin{aligned}
 J_A &= BX + B'Y' \\
 K_A &= B'XY' \\
 J_B &= A'X \\
 K_B &= A + XY' \\
 Z &= AXY + BX'Y'
 \end{aligned}$$



Que 5) Design a sequential circuit with two flip flops, A & B and one input X. When $X=0$ state of the circuit remains the same, when $X=1$ circuit passes through the state transition from 00 to 01 to 11 to 10 back to 00 and repeat.

Given

$A, B, X \rightarrow$ Input

$X=0$ same state

$X=1$ $00 \rightarrow 01 \rightarrow 11 \rightarrow 10 \rightarrow 00$

Let us design using T FF

0	0	0
0	1	1
1	0	1
1	1	0

Present State		Input	Next State		FlipFlop Inputs	
A	B	X	A ₀	B ₀	T _A	T _B
0	0	0	0	0	0	0
0	0	1	0	1	0	1
0	1	0	0	1	0	0
0	1	1	1	1	1	0
1	0	0	1	0	0	0
1	0	1	0	0	1	0
1	1	0	1	1	0	0
1	1	1	1	0	0	1

Present State		Input	Next State		FlipFlop Inputs	
A	B	X	A	B ₀	T _A	T _B
0	0	0	0	0	0	0
0	0	1	0	1	0	1
0	1	0	0	1	0	0
0	1	1	1	1	1	0
1	0	0	1	0	0	0
1	0	1	0	0	1	0
1	1	0	1	1	0	0
1	1	1	1	0	0	1

Kmap for T_A

X \ AB	AB			
	00	01	11	10
0				
1		1		1

$\bar{A}B X$

$A\bar{B} X$

$$\begin{aligned}
 T_A &= \bar{A}B X + A\bar{B} X \\
 &= X (\bar{A}B + A\bar{B}) \\
 &= X (A \oplus B)
 \end{aligned}$$

Kmap for T_B

X \ AB	AB			
	00	01	11	10
0				
1	1		1	

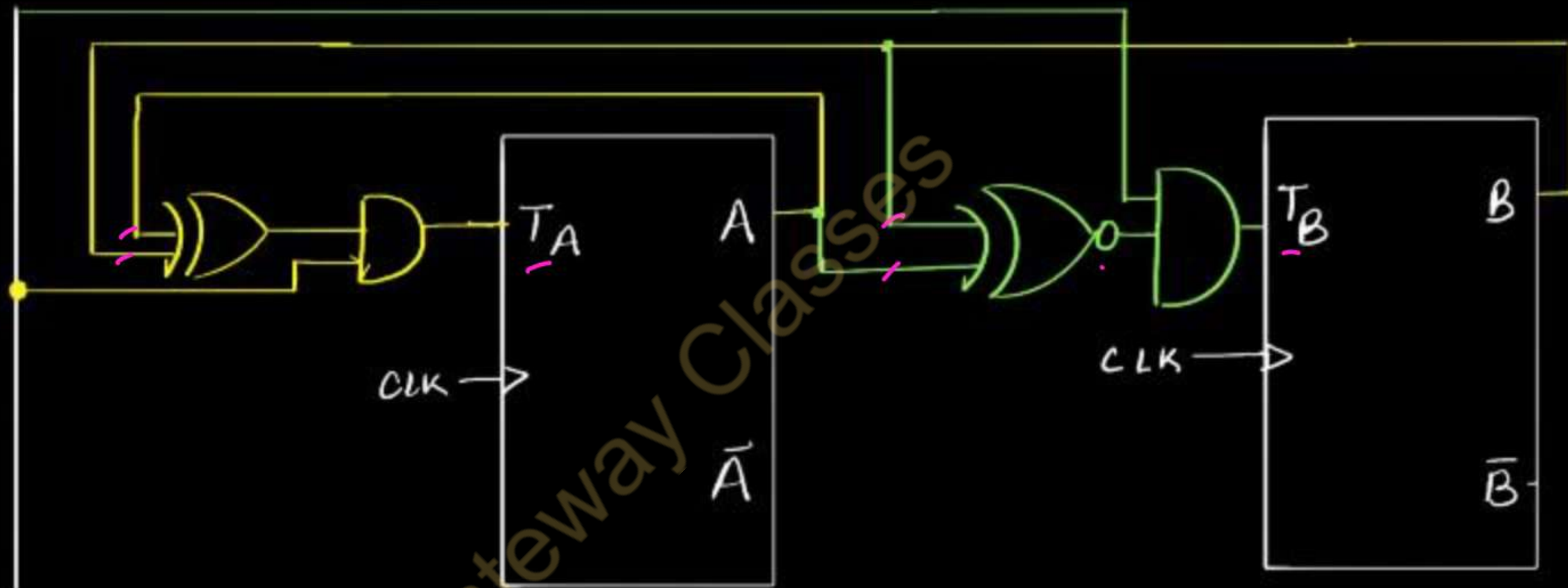
$\bar{A}\bar{B} X$

$AB X$

$$\begin{aligned}
 T_B &= AB X + \bar{A}\bar{B} X \\
 &= X (AB + \bar{A}\bar{B}) \\
 &= X (A \odot B)
 \end{aligned}$$

✓
X

$$T_A = X(A \oplus B)$$
$$\bar{T}_B = X(A \odot B)$$

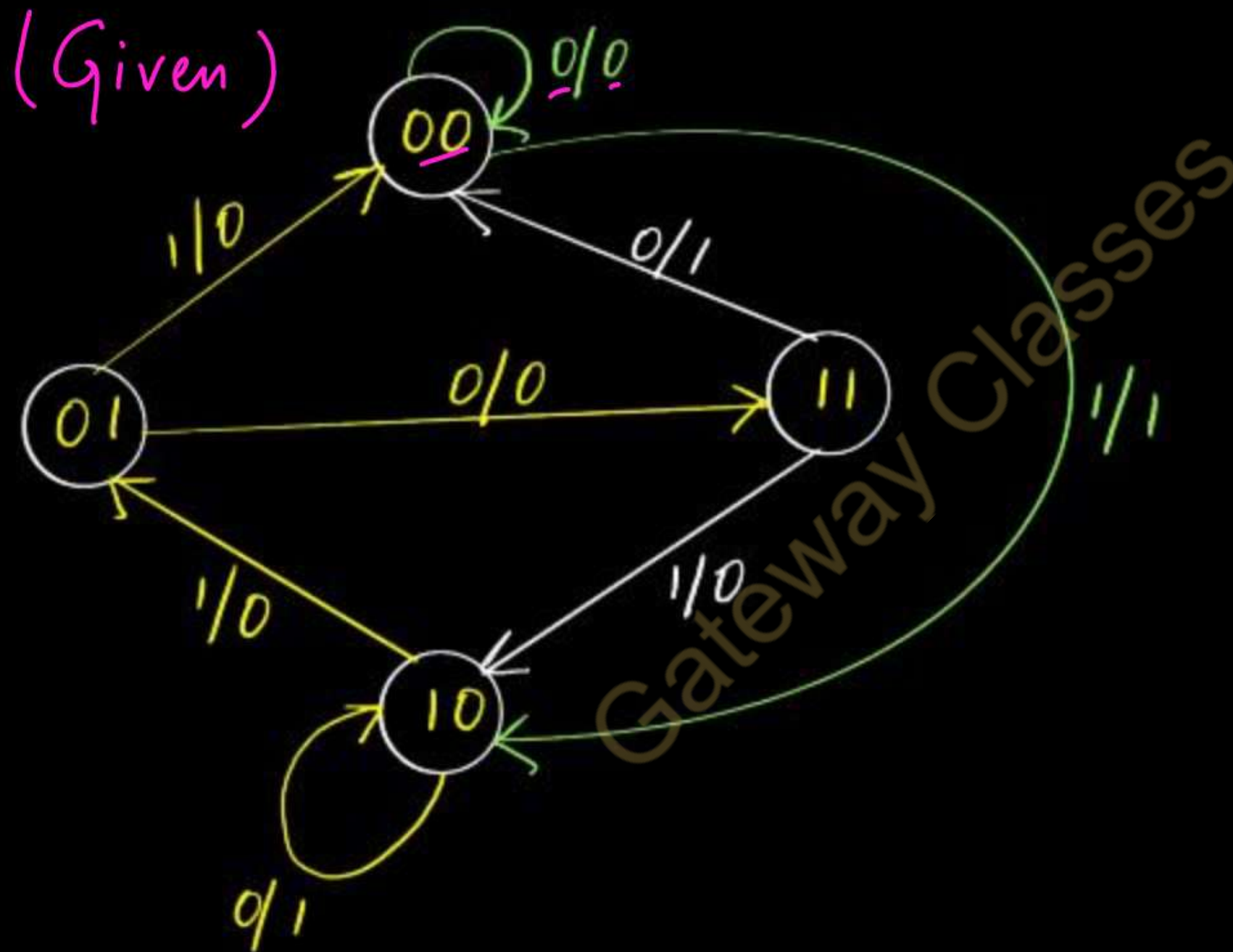


Design Procedure for Clocked (Synchronous) Sequential Circuits

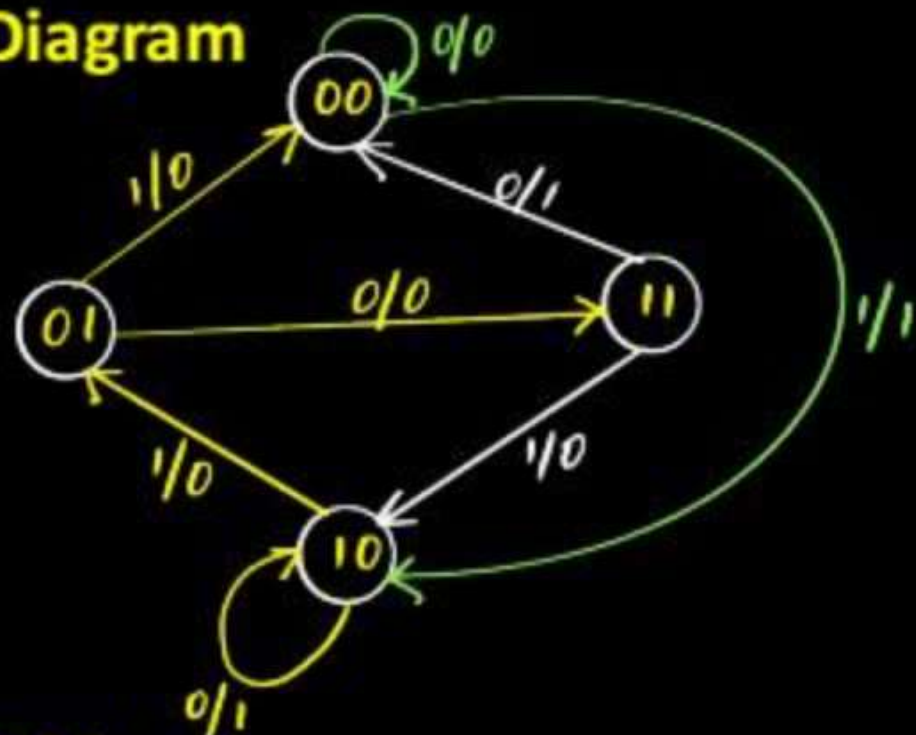
Steps:-

1. State Diagram (Given)
2. From State Diagram make State Table.
3. State Reduction (if applicable)
4. Binary Assignment (if applicable)
5. Find number of flipflops and assign letter (A,B,C,D) to it.
6. Choose type of flipflop [SR , JK , D , T]
7. Compare Excitation Table & State Table and make Circuit Excitation Table.
8. By K-Map, find circuit output flipflop input function.
9. Draw the logic diagram.

Que 6) A sequential circuit has one input and one output. The state diagram is shown below. Design using JK Flipflop.



Step 1 State Diagram



Step 2 State Table

Present State		Next State				Output	
		<u>X=0</u> ¹		<u>X=1</u> ³		<u>X=0</u> ²	<u>X=1</u> ⁴
A	B	A	B	A	B	Y	Y
<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>
<u>0</u>	<u>1</u>	<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>
<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>1</u>	<u>0</u>
<u>1</u>	<u>1</u>	<u>0</u>	<u>0</u>	<u>1</u>	<u>0</u>	<u>1</u>	<u>0</u>

Step 3 & Step 4 are not applicable here as there is no state reduction.

Step 5 Finding the number of Flipflops required.

$$2^n = N$$

$$\text{Here, } 2^n = 4$$

$$2^n = 2^2$$

So $n = 2$, 2 FFs are required. (A, B)

Step 6 Flipflop is JK (specified in Question)

Step 7 Excitation Table for JK FF

Q _n	Q _{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

$\nearrow J_A \quad K_A$
 $\searrow J_B \quad K_B$

State Table

Present State		Present State		Next State		Output	
				X=0	X=1	X=0	X=1
A	B	A	B	A	B	Y	Y
0	0	0	0	0	0	0	1
0	1	0	1	1	1	0	0
1	0	1	0	1	0	1	0
1	1	1	1	0	0	1	0

Excitation Table for JK FF

Q _n	Q _{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Circuit Excitation Table

Present State		Input	Next State		Flipflop Inputs				Output
A	B	X	A	B	J _A	K _A	J _B	K _B	Y
0	0	0	0	0	0	X	0	X	0
0	0	1	1	0	1	X	0	X	1
0	1	0	1	1	1	X	X	0	0
0	1	1	0	0	0	X	X	1	0
1	0	0	1	0	X	0	0	X	1
1	0	1	0	1	X	1	1	X	0
1	1	0	0	0	X	1	X	1	1
1	1	1	1	0	X	0	X	1	0

see A, A

see B, B

Step 8 K Maps

For J_A

$x \backslash AB$	00	01	11	10
0		1	X	x
1	1		X	X

Annotations: $B\bar{X}$ (pointing to cell 01, 0) and $\bar{B}X$ (pointing to cell 00, 1).

$$J_A = B\bar{X} + \bar{B}X$$

$$J_A = B \oplus X$$

For J_B

$x \backslash AB$	00	01	11	10
0		X	X	
1		X	X	1

Annotation: AX (pointing to cell 11, 1).

$$J_B = AX$$

For Output Y

$x \backslash AB$	00	01	11	10
0			1	1
1	1			

Annotations: $\bar{A}\bar{B}X$ (pointing to cell 00, 1) and $A\bar{X}$ (pointing to cell 11, 0).

$$Y = \bar{A}\bar{B}X + A\bar{X}$$

For K_A

$x \backslash AB$	00	01	11	10
0	X	X	1	
1	X	X		1

Annotations: $B\bar{X}$ (pointing to cell 01, 0) and $\bar{B}X$ (pointing to cell 10, 1).

$$K_A = B\bar{X} + \bar{B}X$$

$$= B \oplus X$$

For K_B

$x \backslash AB$	00	01	11	10
0	X		1	X
1	X	1	1	X

Annotations: X (pointing to cell 01, 1) and A (pointing to cell 11, 0).

$$K_B = A + X$$

Step 9 Logic Diagram

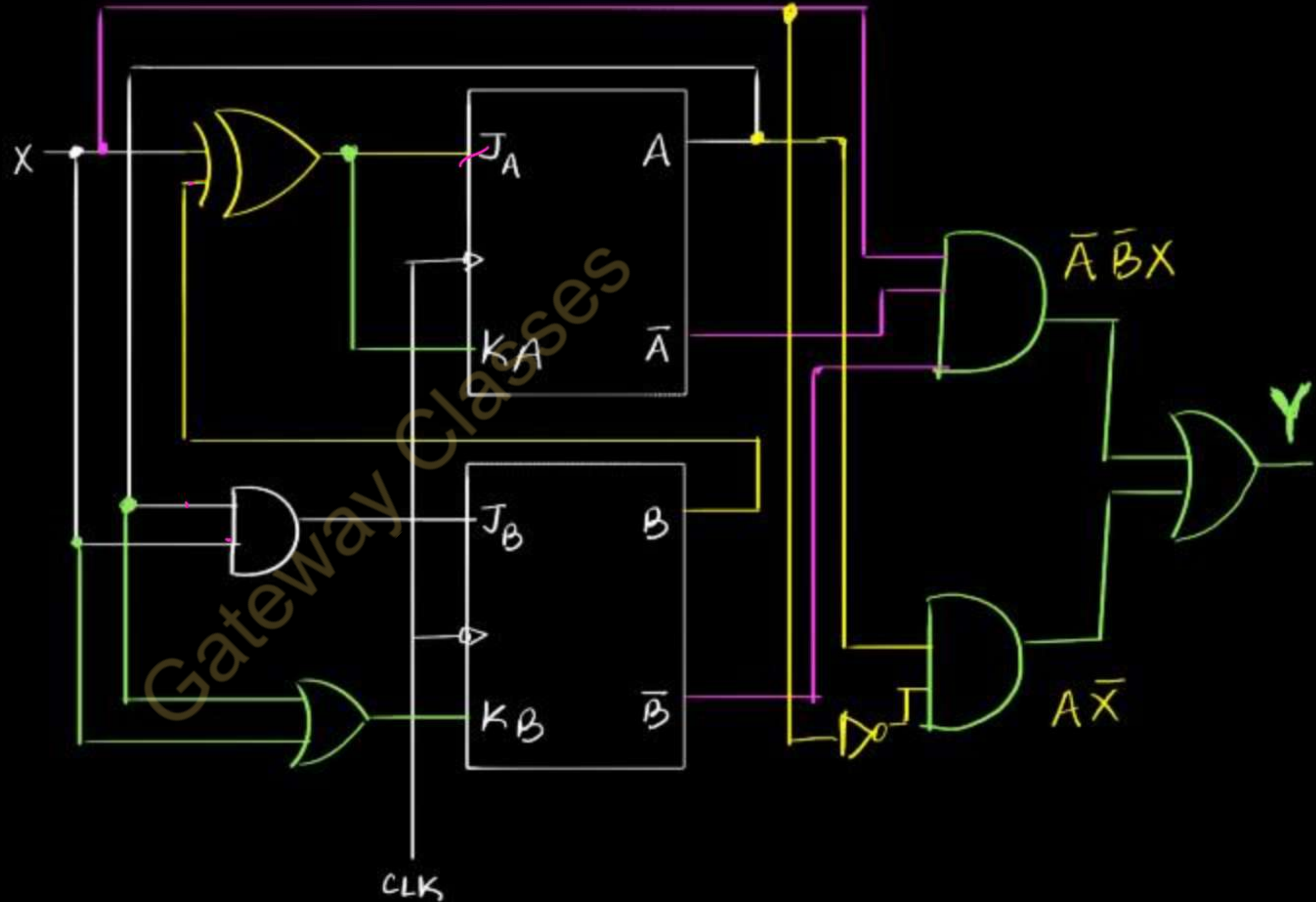
$$J_A = B \oplus X \quad \checkmark$$

$$J_B = AX$$

$$K_A = B \oplus X \quad \checkmark$$

$$K_B = A + X$$

$$Y = \bar{A}\bar{B}X + A\bar{X}$$



Design Procedure for Asynchronous Sequential Circuits

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Steps:-

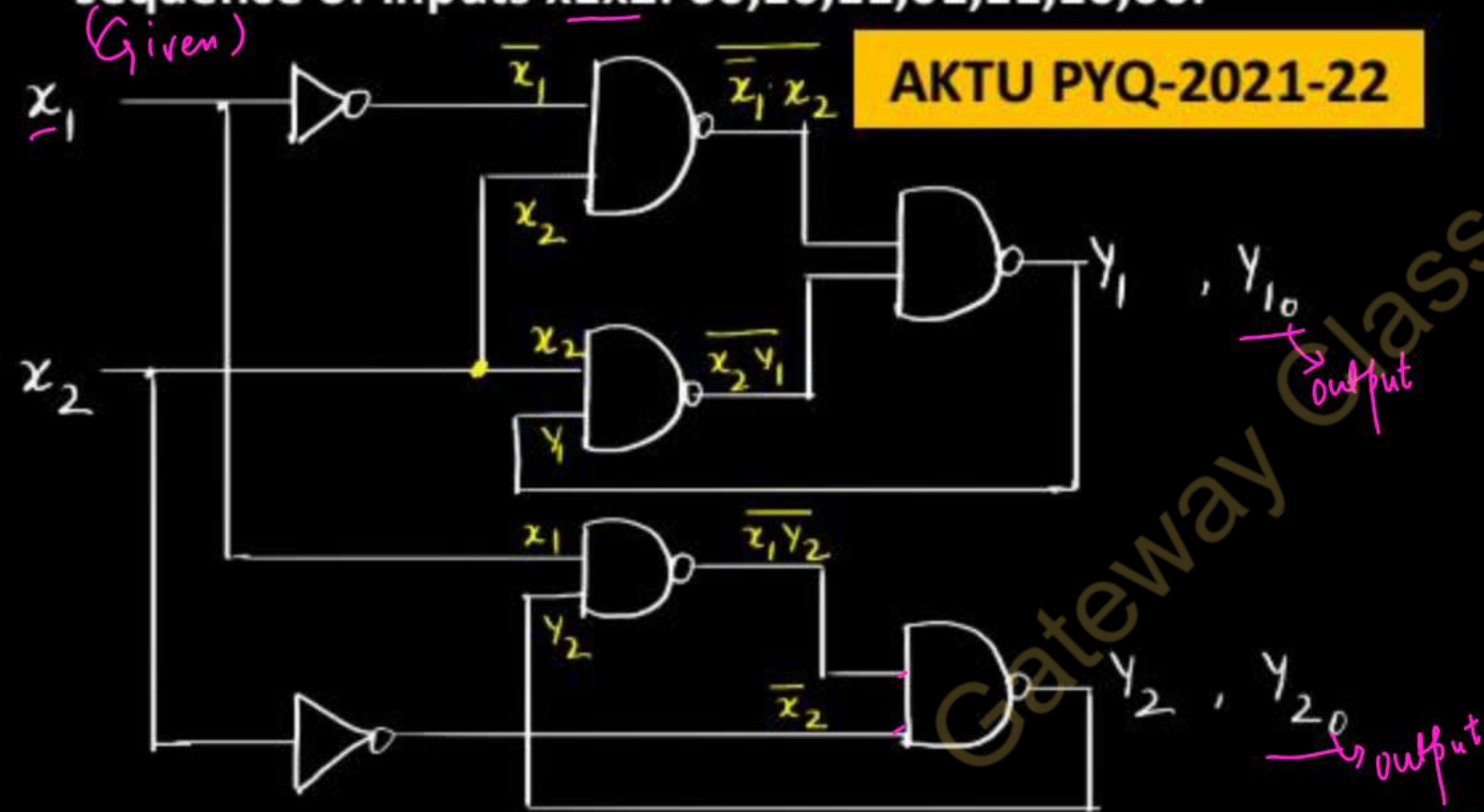
1. State Diagram and Primitive FlowTable (*is a primary tool for designing and analyzing state transitions*)
2. Reduction of Primitive FlowTable
3. Transition Table and State Assignment (ensuring a race-free assignment i.e no critical races during state changes)
4. Output Map
5. Derive Boolean Equations
6. Simplify and Realize using K-Map
7. Draw Logic diagram

Explain the role of flow table in designing asynchronous sequential circuits.

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Que 7) Derive the **transition table** for the asynchronous sequential circuit shown in given figure. Determine the sequence of internal states Y_1 Y_2 for the following sequence of inputs x_1x_2 : 00,10,11,01,11,10,00.

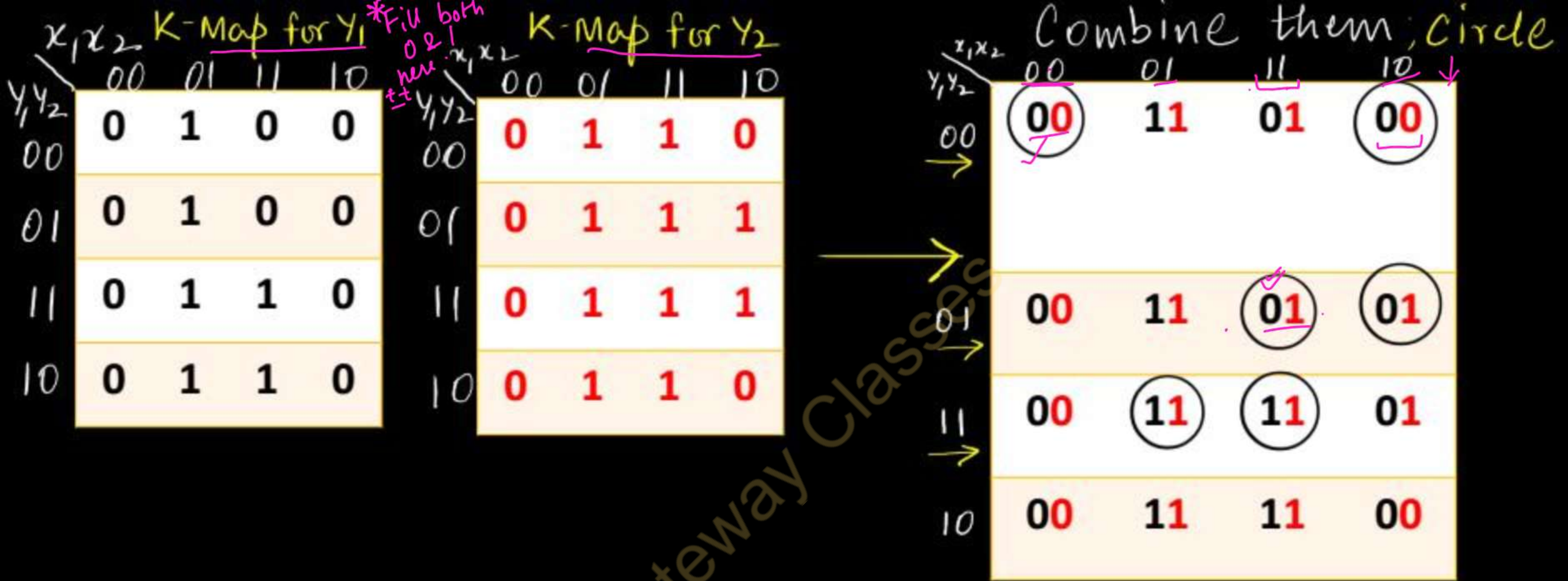
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$$y_{1o} = \overline{\overline{x_1 x_2} \cdot \overline{x_2 y_1}} = \overline{\overline{x_1 x_2}} + \overline{\overline{x_2 y_1}} = x_1 x_2 + x_2 y_1 \quad (1)$$

$$y_{2o} = \overline{\overline{x_1 y_2} \cdot \overline{x_2}} = \overline{\overline{x_1 y_2}} + \overline{\overline{x_2}} = x_1 y_2 + x_2 \quad (2)$$

x_1	x_2	Y_1	Y_2	Y_{1o}	Y_{2o}
0	0	0	0	0	0
0	0	0	1	0	0
0	0	1	0	0	0
0	0	1	1	0	0
0	1	0	0	1	1
0	1	0	1	1	1
0	1	1	0	1	1
0	1	1	1	1	1
1	0	0	0	0	0
1	0	0	1	0	1
1	0	1	0	0	0
1	0	1	1	0	1
1	1	0	0	0	1
1	1	0	1	0	1
1	1	1	0	1	1
1	1	1	1	1	1



Transition Table

x_1x_2	00	10	11	01	11	10	00
Y_1Y_2	00	00	01 ✓	11	11	01	00

Ans

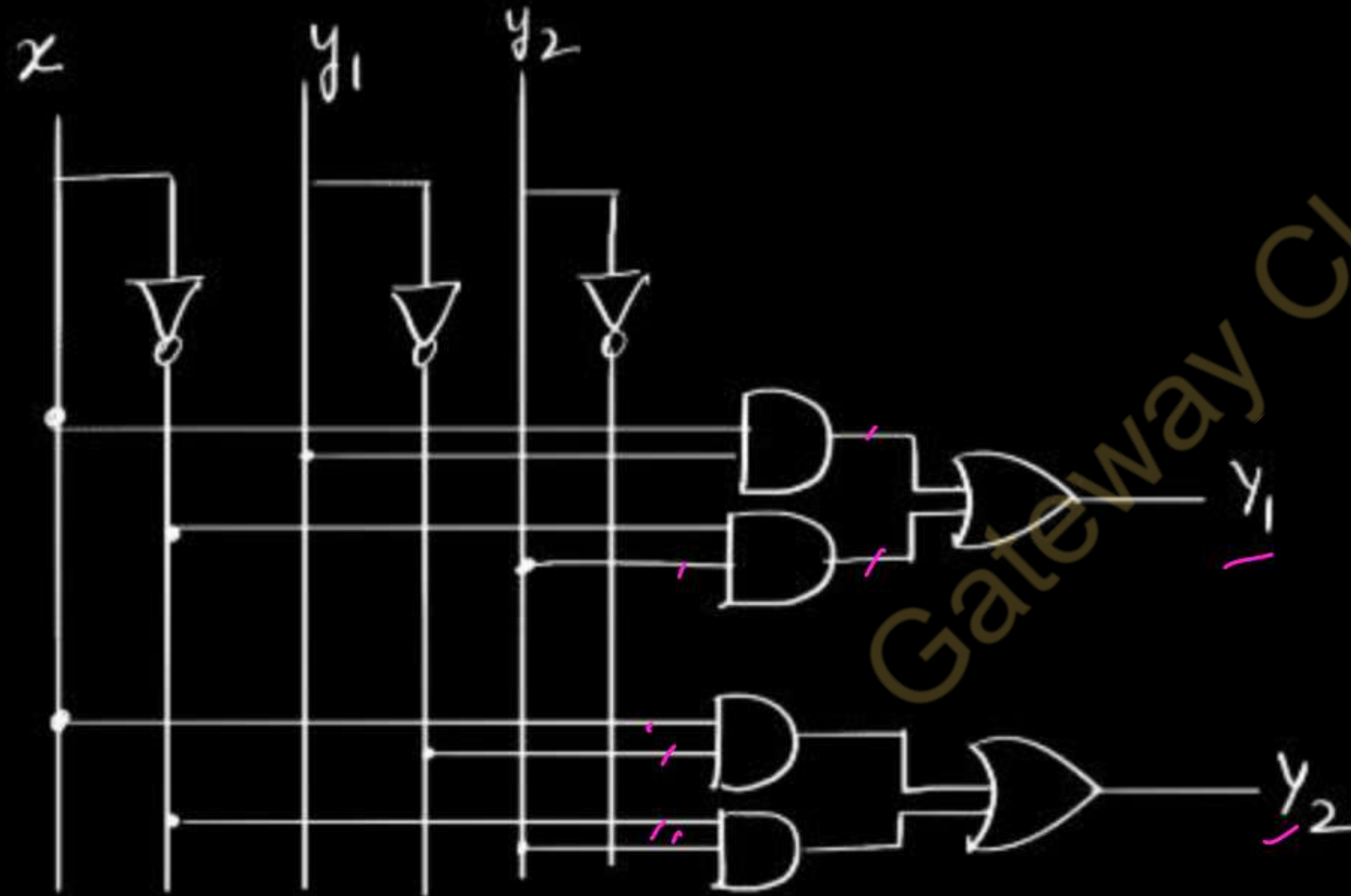
Que 8) An asynchronous sequential circuit with two excitation function with two feedback loop is given as :

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✓ $Y_1 = xy_1 + x'y_2$; ✓ $Y_2 = xy_1' + x'y_2$

(i) Draw the logic diagram of the circuit.

(ii) Derive the transition table & obtain the flow table.



$Y_1 = xy_1 + x'y_2$; $Y_2 = xy_1' + x'y_2$

x	y ₁	y ₂	Y ₁	Y ₂
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	1	0
1	1	1	1	0

x	y ₁	y ₂	Y ₁	Y ₂
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	1	0
1	1	1	1	0

For y₁

x	0	1
y ₁ y ₂	00	00
01	1	0
11	1	1
10	0	1

For y₂

x	0	1
y ₁ y ₂	00	01
01	1	1
11	1	0
10	0	0

Combine them, Circle

x	0	1
y ₁ y ₂	00	01
01	11	01
11	11	10
10	00	10

State Assignment

Let a -> 00

b -> 01

c -> 11

d -> 10

	0	1
a	a	b
b	c	b
c	c	d
d	a	d

(Transition Table)

(Flow Table)

~~imp~~ Que 9) Design a **Primitive State Diagram** and **Primitive State Table** of an Asynchronous Sequential Circuit with two inputs x1 and x2 and output z.

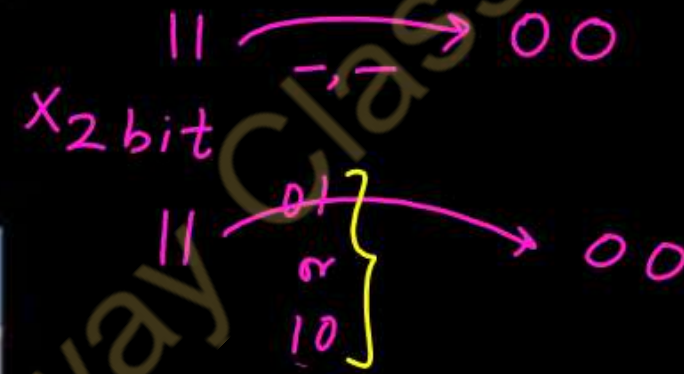
Initially both the inputs are equal to zero.

When x1 or x2 becomes 1 the output becomes 1. Row 3, Row 2 → o/p 1

When the second input also becomes 1, the output changes to 0. Row 4, o/p 0

The output stays at 0 until the circuit goes back to initial state.

X1	X2	Output Z
0	0	0
0	1	1
1	0	1
1	1	0



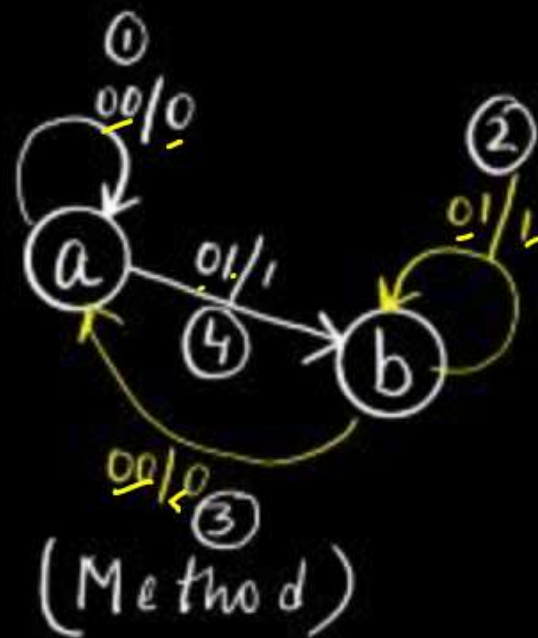
X1	X2	Output Z
a → 0	0	0
b → 0	1	1
c → 1	0	1
d → 1	1	0

e 0
f 1
0 0

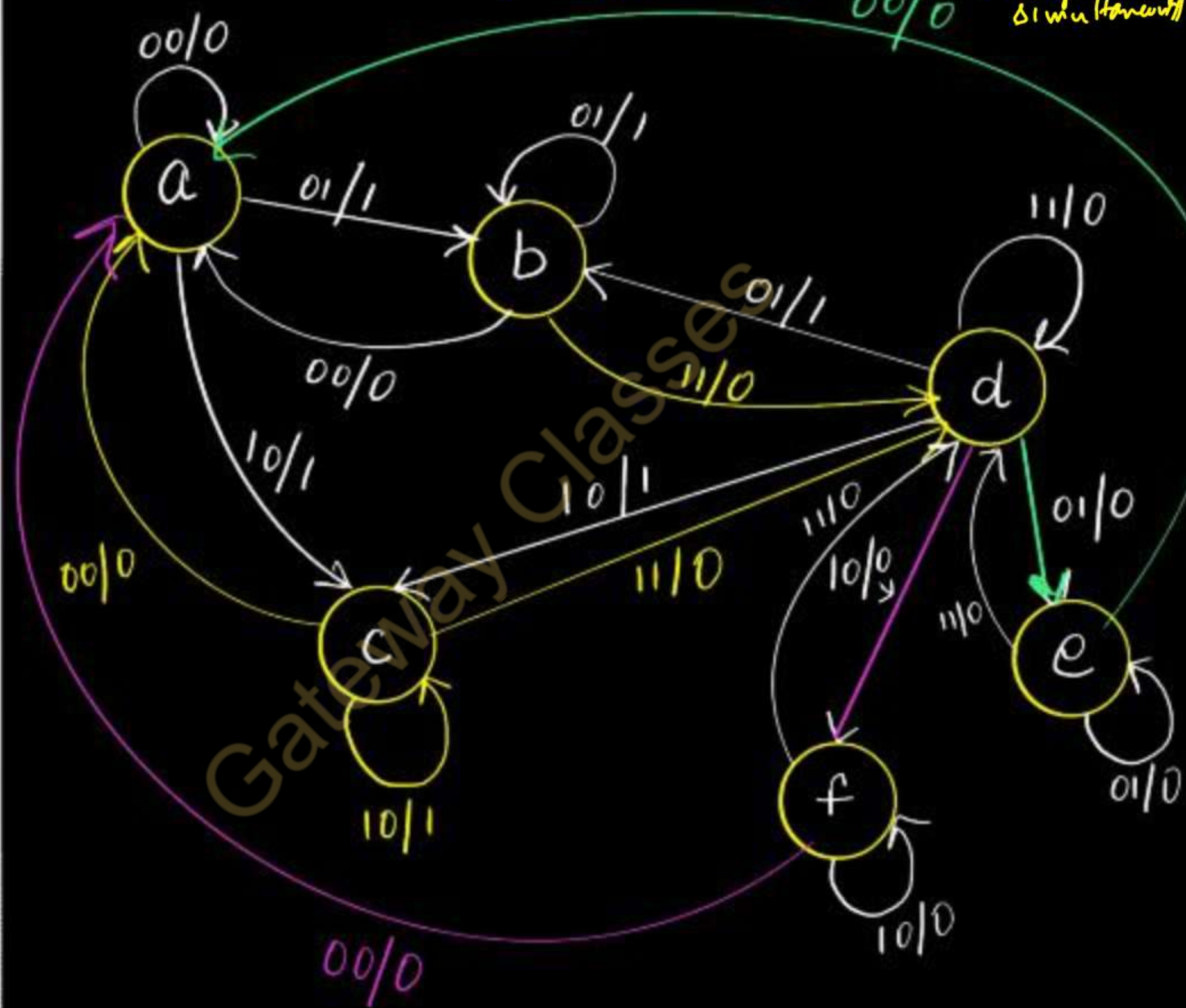
Notation

x1 x2 / Z
of destination

Assume
intermediate
status bit
* d → a



Primitive State Diagram → (Notation change, Two bits cannot change simultaneously)



- a ① a → a
b ② b → b
③ b → a
④ a → b

c → c
c → a
c → b x
a → c

d → d
d → a x
d → b
d → c
b → d, c → d

e * d → a
(i) d → e → a
01

f (ii) d → f → a
10

e → e
f → f

e → d
f → d

Primitive State Table

Points to remember

Single bit change is allowed

- 00 → 00
- 00 → 01
- 00 → 10
- 00 → 11 X
- 01 → 00
- 01 → 01
- 01 → 11
- 01 10 X
- 10 00
- 10 01 X
- 10 10
- 10 11
- 11 00 X
- 11 01
- 11 10
- 11 11

a - 00

b - 01

c - 10

d - 11

e - 01

f - 10

To fill in Table

State, output

Circle Stable State (red marked here)

a → a

b → b

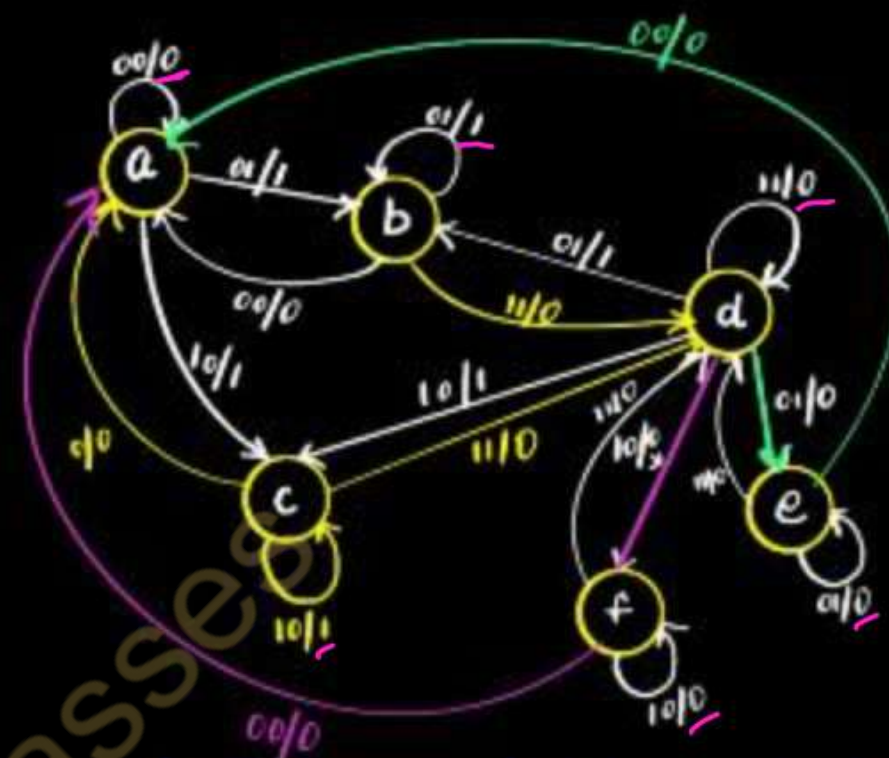
c → c

d → d

e → e

f → f

Only one stable state in a row.
Output of only stable state is written, for unstable states - is written.



PS	00 a	01 b	11 d	10 c
a 00	a, 0	b, -	-, -	c, -
b 01	a, -	b, 1	d, -	-, -
c 10	a, -	-, -	d, -	c, 1
d 11	-, -	e, -	d, 0	f, -
e 01	a, -	e, 0	d, -	-, -
f 10	a, -	-, -	d, -	f, 0

Additionally, we can design the circuit

Ans.

PS	00 a	01 b	11 d	10 c
a 00	a, 0	b, -	- , -	c, -
b 01	a, -	b, 1	d, -	- , -
c 10	a, -	- , -	d, -	c, 1
d 11	- , -	e, -	d, 0	f, -
e 01	a, -	e, 0	d, -	- , -
f 10	a, -	- , -	d, -	f, 0

Merge 3 -3 rows

PS	00 a	01 b	11 d	10 c
<u>a,b,c</u>	a, 0	b, 1	d, -	c, 1
<u>d,e,f</u>	a, -	e, 0	d, 0	f, 0

Replace a,b,c by a and d,e,f by d

PS	00 a	01 b	11 d	10 c
a	a, 0	a, 1	d, -	a, 1
d	a, -	d, 0	d, 0	d, 0

Let a $\rightarrow$ 0 , d $\rightarrow$ 1

PS	00 a	01 b	11 d	10 c
0	0, 0	0, 1	1, -	0, 1
1	0, -	1, 0	1, 0	1, 0

M
R
A
K-Map
Logic diag.

Designing Using D FF

K Map for input D

x_1, x_2	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$x_2 y$ (for 01, 11)
 $x_1 x_2$ (for 11, 10)
 $x_1 y$ (for 11, 10)

K Map for output z

x_1, x_2	00	01	11	10
0	0	1	0	1
1	0	0	0	0

$\bar{x}_1 x_2 \bar{y}$ (for 01)
 $x_1 \bar{x}_2 \bar{y}$ (for 10)

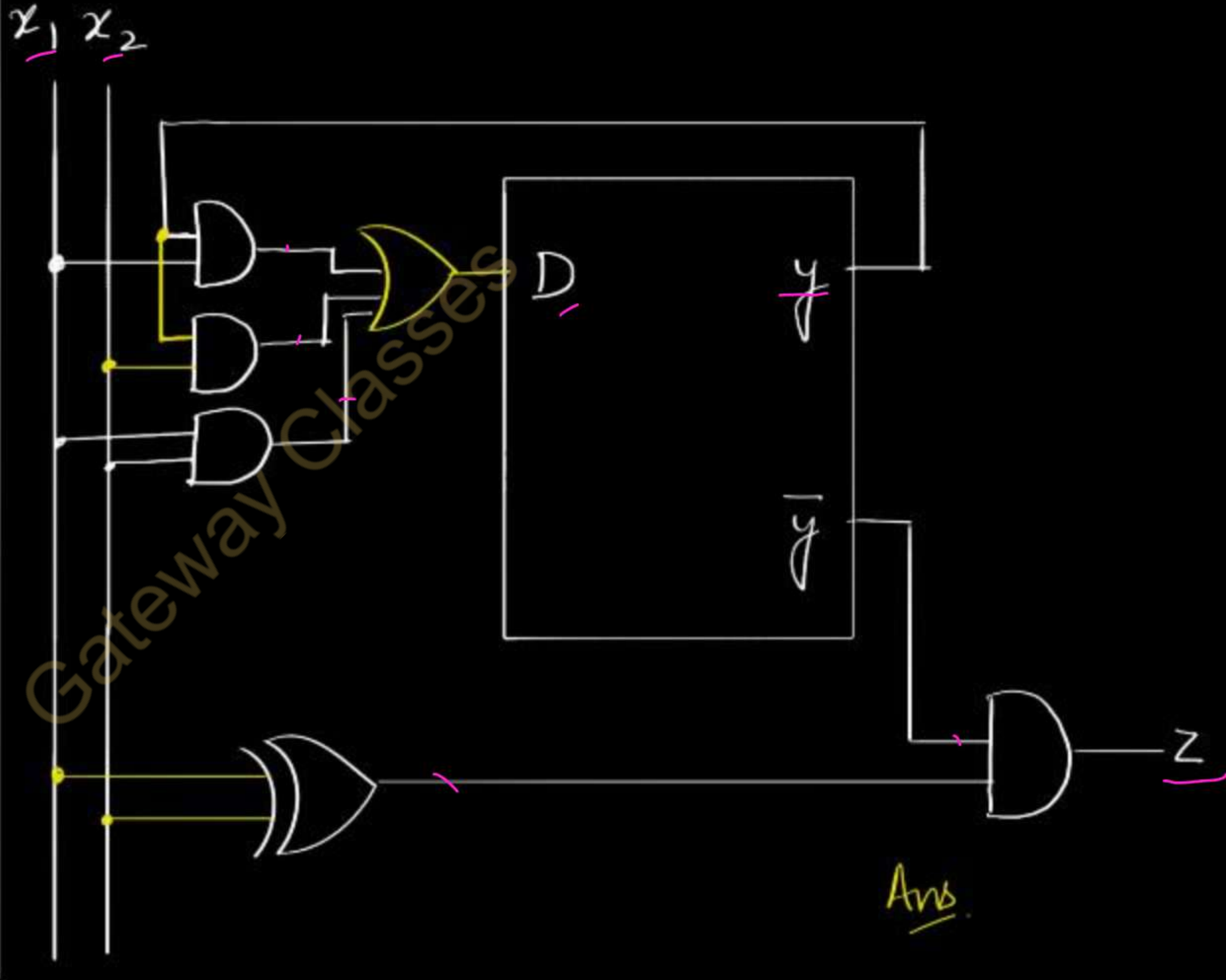
$$D = x_1 x_2 + x_1 y + x_2 y$$

$$\begin{aligned}
 z &= \bar{x}_1 x_2 \bar{y} + x_1 \bar{x}_2 \bar{y} \\
 &= \bar{y} (x_1 \bar{x}_2 + \bar{x}_1 x_2) \\
 z &= \bar{y} (x_1 \oplus x_2)
 \end{aligned}$$

Logic Diagram

$$D = x_1 x_2 + x_1 y + x_2 y$$

$$Z = \bar{y} (x_1 \oplus x_2)$$



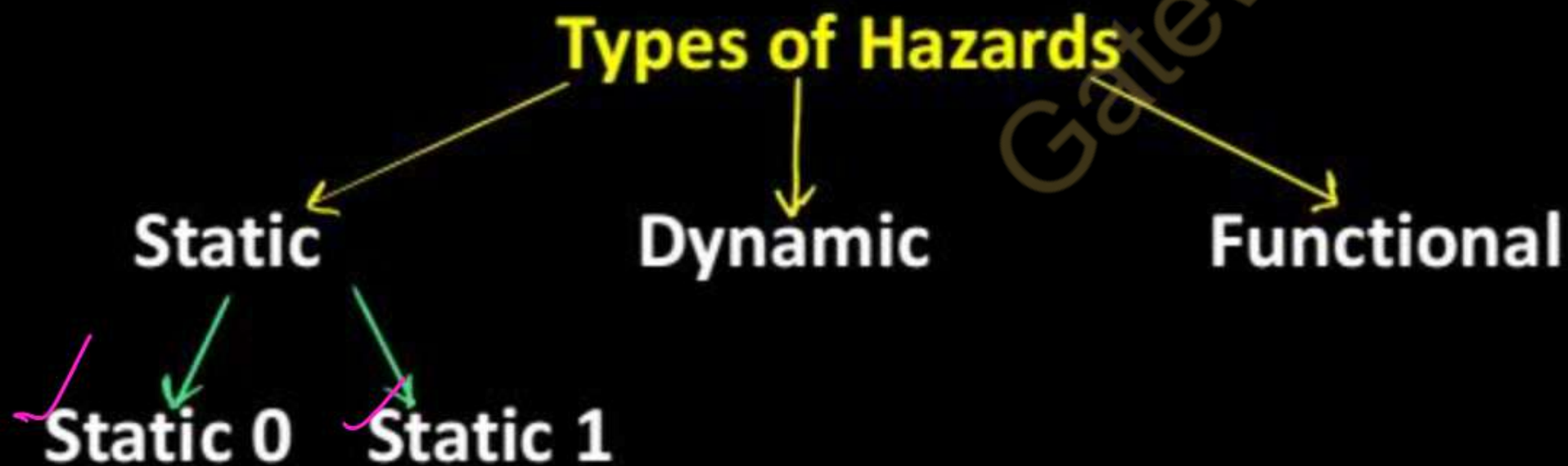
Hazards

AKTU PYQ-2022-23,2021-22

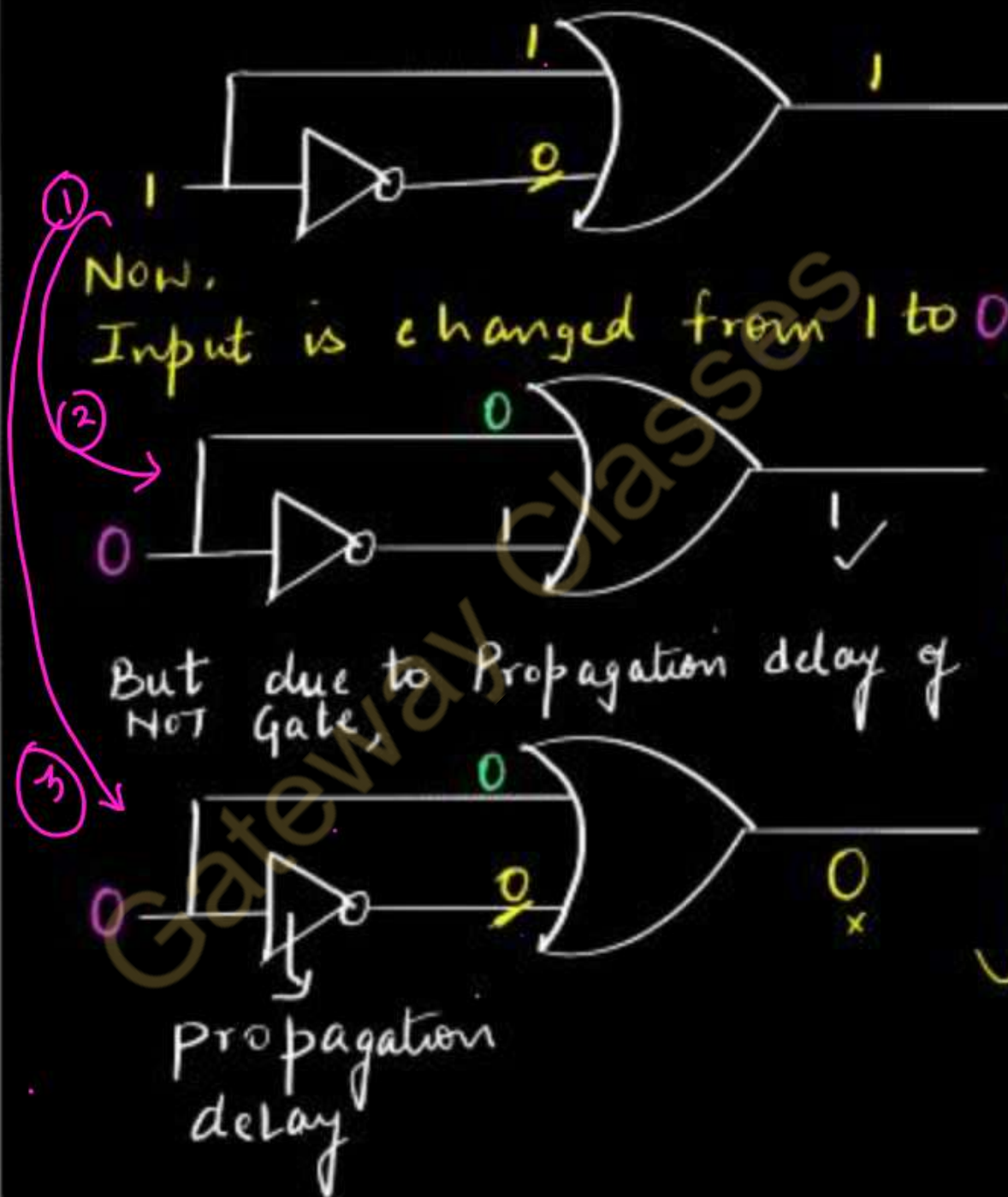
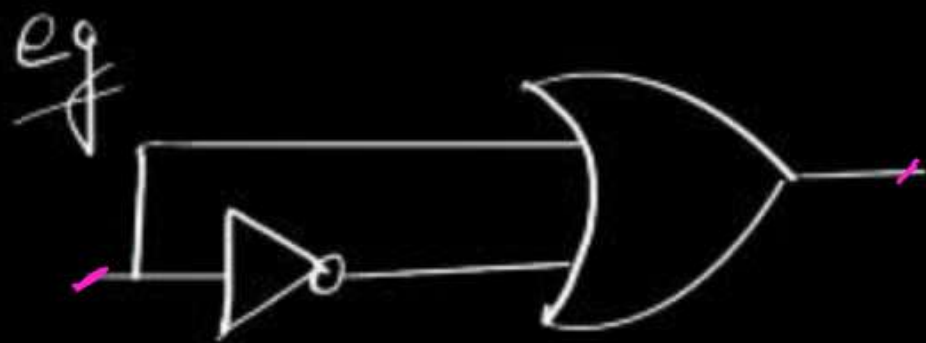
Hazards are **temporary, unwanted switching transitions** at the output of a circuit.

These transitions are also known as signal **Glitch (spike)**

It occurs due to Different paths of Logic circuits and different **propagation delays** in signal paths, leading to brief **incorrect logic states (0 or 1)** before settling to the correct output.



Hazards



Output should be 1 but it is 0.

Static-1 Hazard

It occurs for ms, and it is auto-corrected.

But problem occurs if it is propagated further in the circuit.

Extra Gates are added.

Discuss hazards in asynchronous circuits. How can race conditions be avoided?

AKTU PYQ-2024-25

Varying Propagation Delays: Different logic gates (AND, OR, NOT) and wire lengths cause signals to arrive at different times.

Multiple Paths: A single input change travels along several paths within the circuit, and these paths have different delays, creating a "race condition".

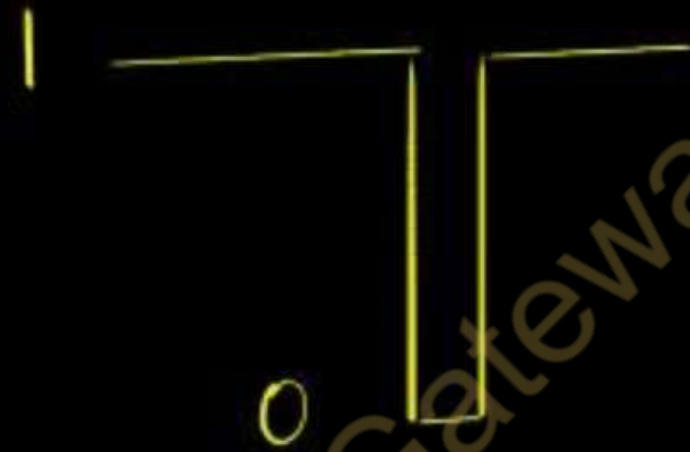
How to Deal with Hazards

Elimination (**Static Hazards**): Add redundant logic (**extra gates**) to cover the "gap" in the **Karnaugh map**, ensuring the term remains true (or false) across the transition.

Avoidance (**Dynamic Hazards**): Use **two-level logic** (**AND-OR** or **NAND-NAND**) and eliminate all static hazards; this inherently prevents dynamic hazards.

Static Hazards: Occurs when an input change causes the output to momentarily change to the *wrong* value before settling to the *correct* value.

Static-1 Hazard: Output momentarily goes to 0 when it should stay 1
(1 $\rightarrow$ 0 $\rightarrow$ 1).



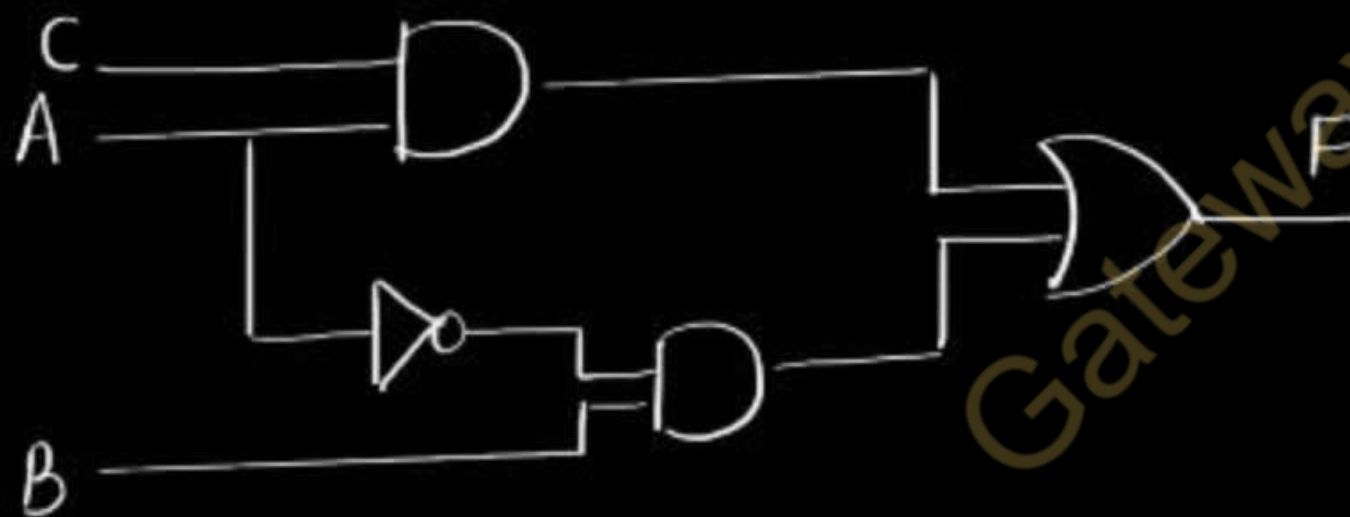
Example of Static 1 Hazard

C \ AB	00	01	11	10
0	0	1	0	0
1	0	1	1	1

$\bar{A}B$

AC

$$F = \bar{A}B + AC$$



Q

$$F = \sum(2, 3, 5, 7)$$

C \ AB	00	01	11	10
0	0	2	6	4
1	1	3	7	5

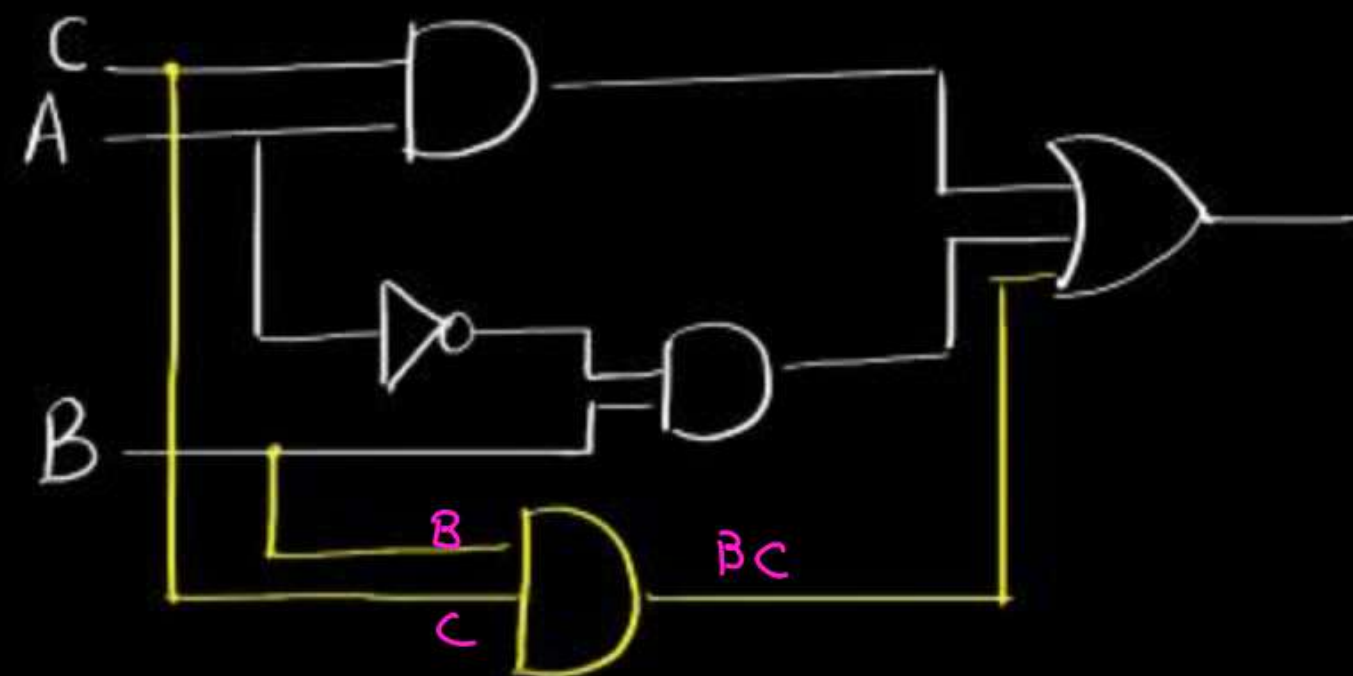
C \ AB	00	01	11	10
0	0	1	0	0
1	0	1	1	1

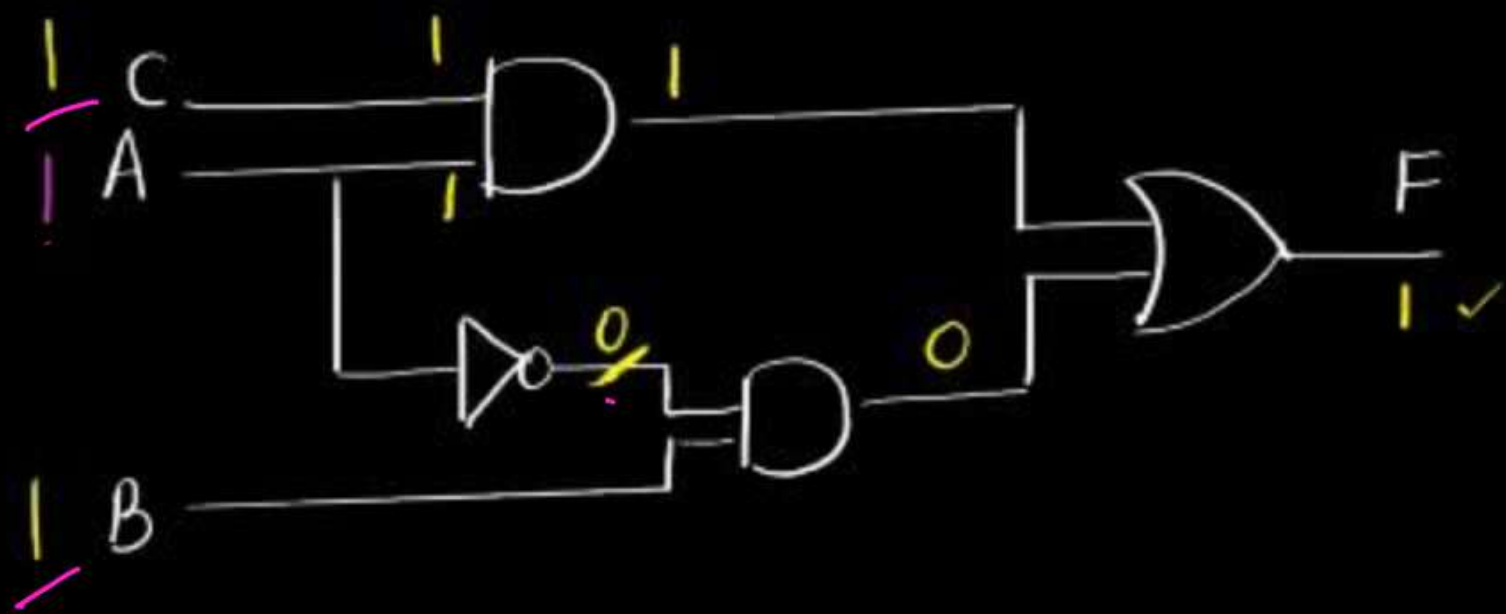
$\bar{A}B$

BC

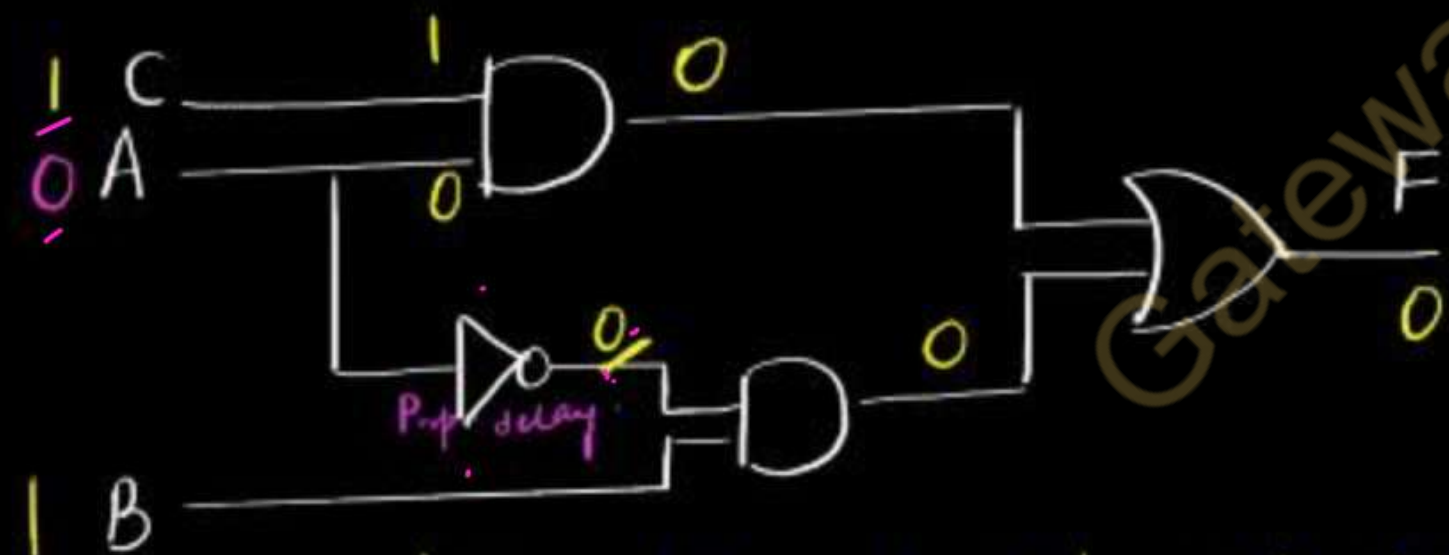
we will get Hazard at $B=C=1$

$$F = \bar{A}B + AC + \underline{BC}$$

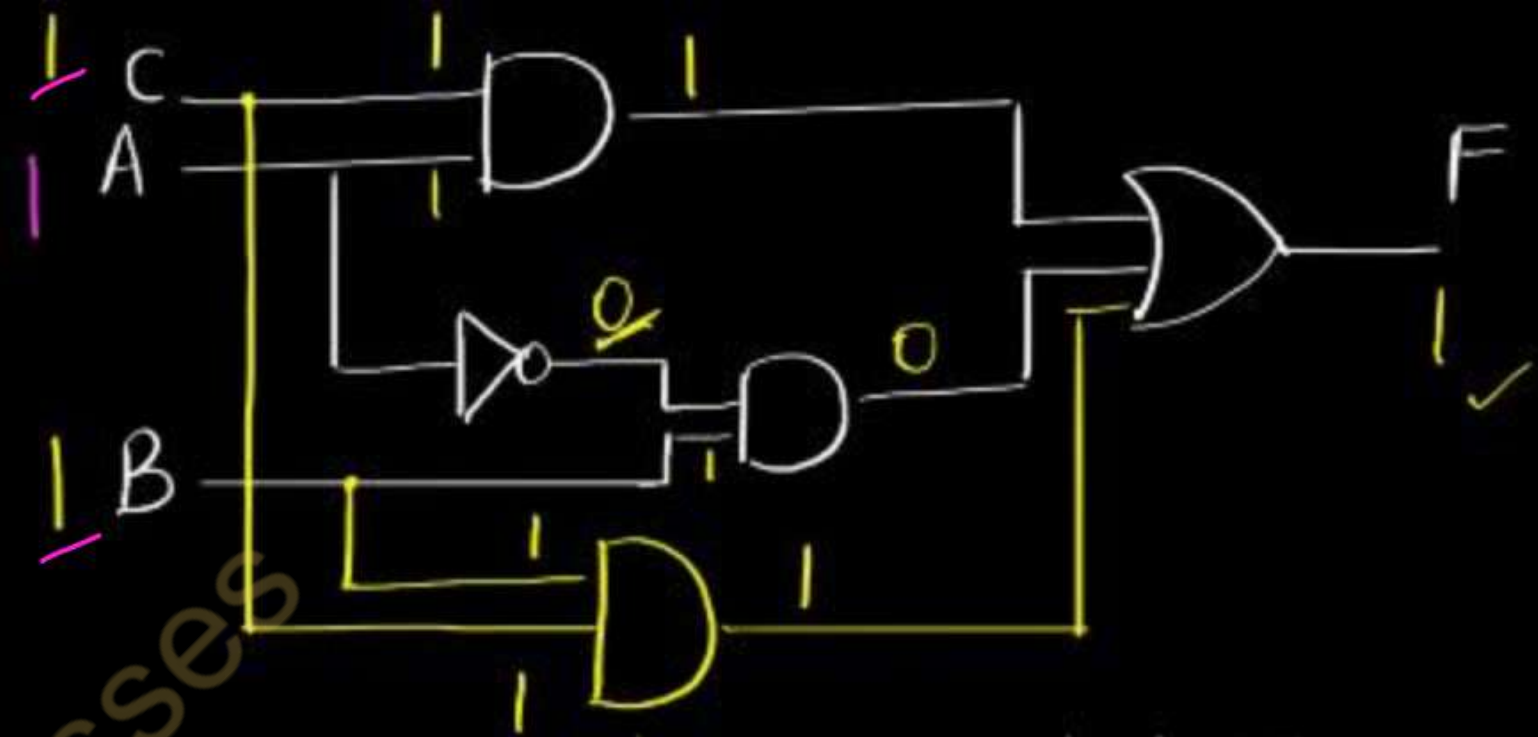




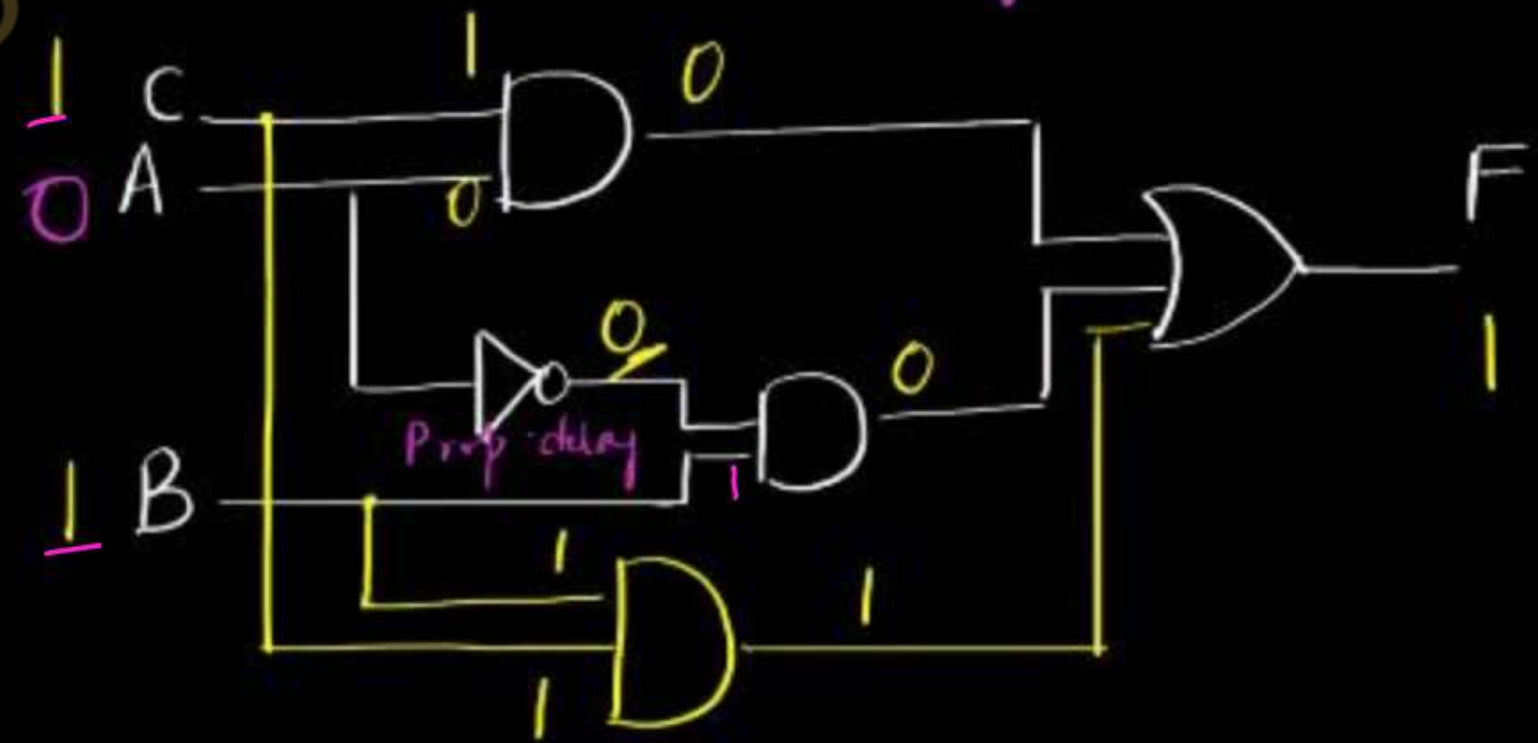
Now Let A is changed to 0,



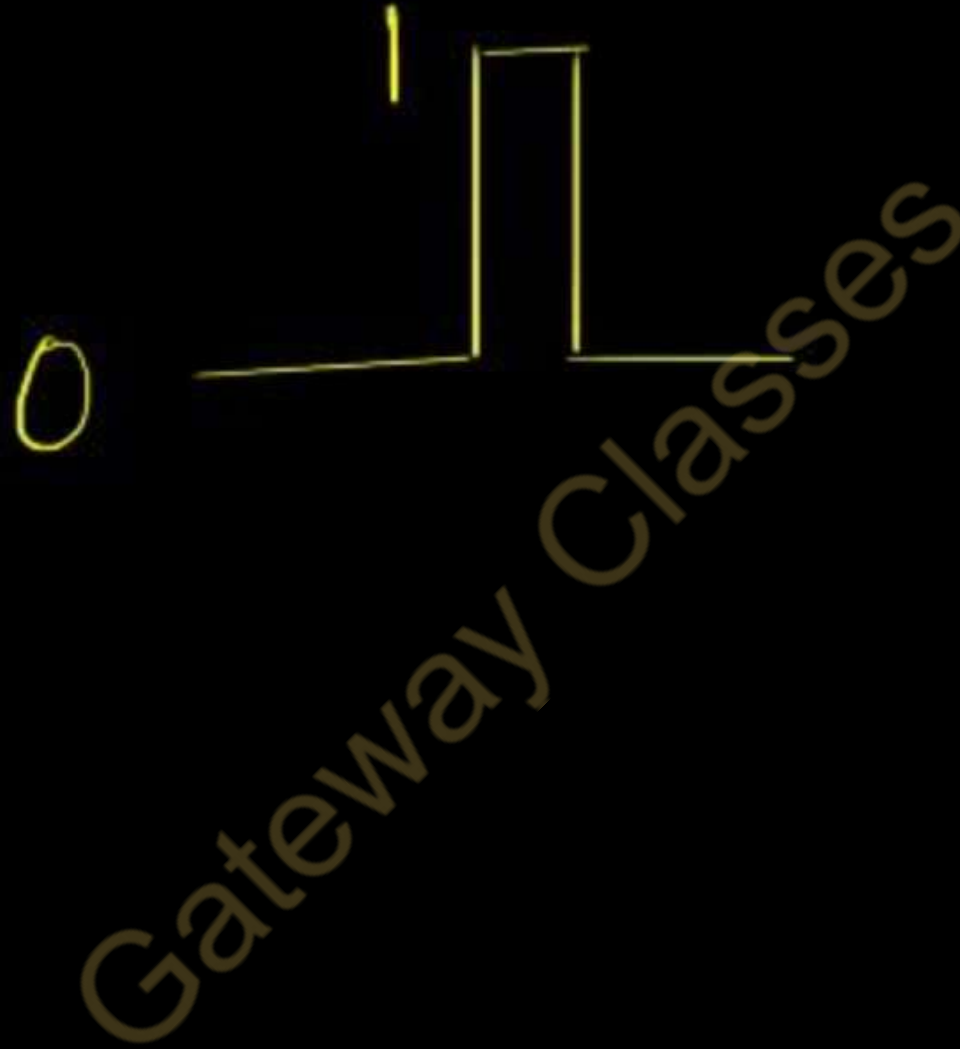
with the help of k-map,
1 AND gate is added for BC .



Now Let A is changed to 0,



Static-0 Hazard: Output momentarily goes to 1 when it should stay 0 (0 $\rightarrow$ 1 $\rightarrow$ 0).



Dynamic Hazard: Occurs during an intended logic transition (0 to 1, or 1 to 0) when the output switches three or more times (e.g., 0 $\rightarrow$ 1 $\rightarrow$ 0 $\rightarrow$ 1).



Distinguish between static and dynamic hazard. How will you determine hazard in combinational circuits?

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Functional Hazards (or Function Hazards) happen with **multiple input changes** and can't be fixed by logic, while

Essential Hazards occur in asynchronous circuits with **single input changes** due to unequal delays in feedback paths and *can* sometimes be corrected by delaying signals

Gateway Classes

What are essential hazards?

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Que10) Draw the logic diagram of the product-of-sums expression

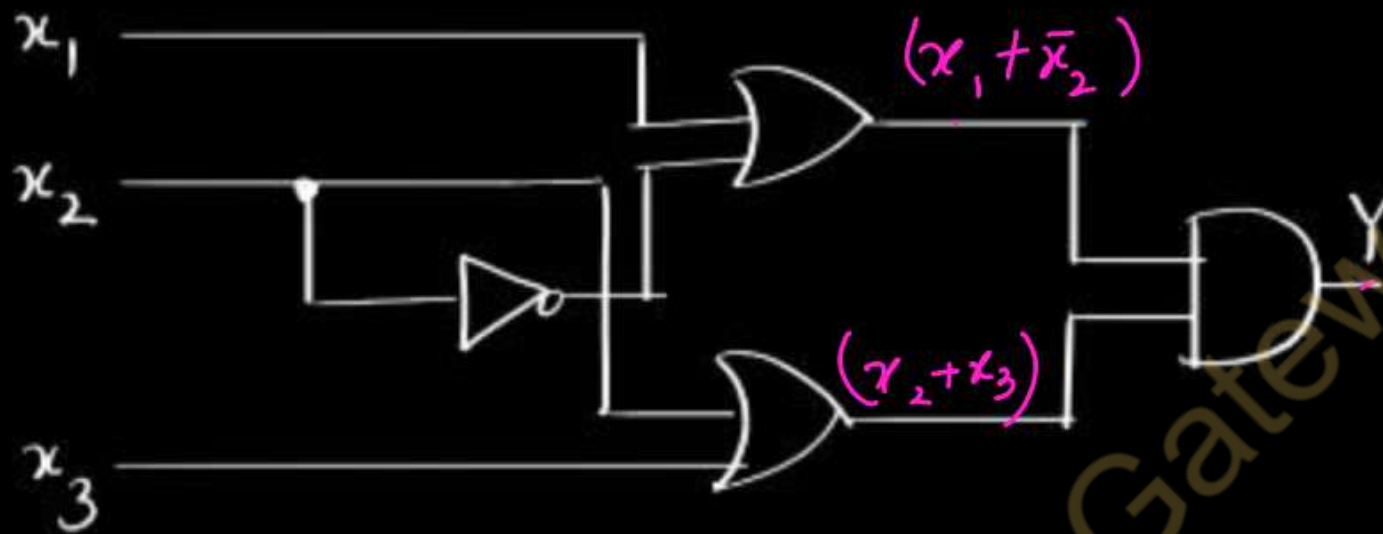
AKTU PYQ-2021-22

$$Y = (x_1 + x_2')(x_2 + x_3).$$

POS.

Show that there is a static 0-hazard when x_1 and x_3 are equal to 0 and x_2 goes from 0 to 1. Find a way to remove the hazard by adding one more OR gate.

$$Y = (x_1 + \bar{x}_2)(x_2 + x_3)$$



Truth Table

x_1	x_2	x_3	$\bar{x}_2$	$x_1 + \bar{x}_2$	$x_2 + x_3$	$(x_1 + \bar{x}_2) \cdot (x_2 + x_3)$
0	0	0	1	1	0	0 ✓
0	0	1	1	1	1	1 ✓
0	1	0	0	0	1	0 ✓
0	1	1	0	0	1	0 ✓
1	0	0	1	1	0	0 ✓
1	0	1	1	1	1	1
1	1	0	0	0	1	0
1	1	1	0	0	1	0

POS from

K-Map

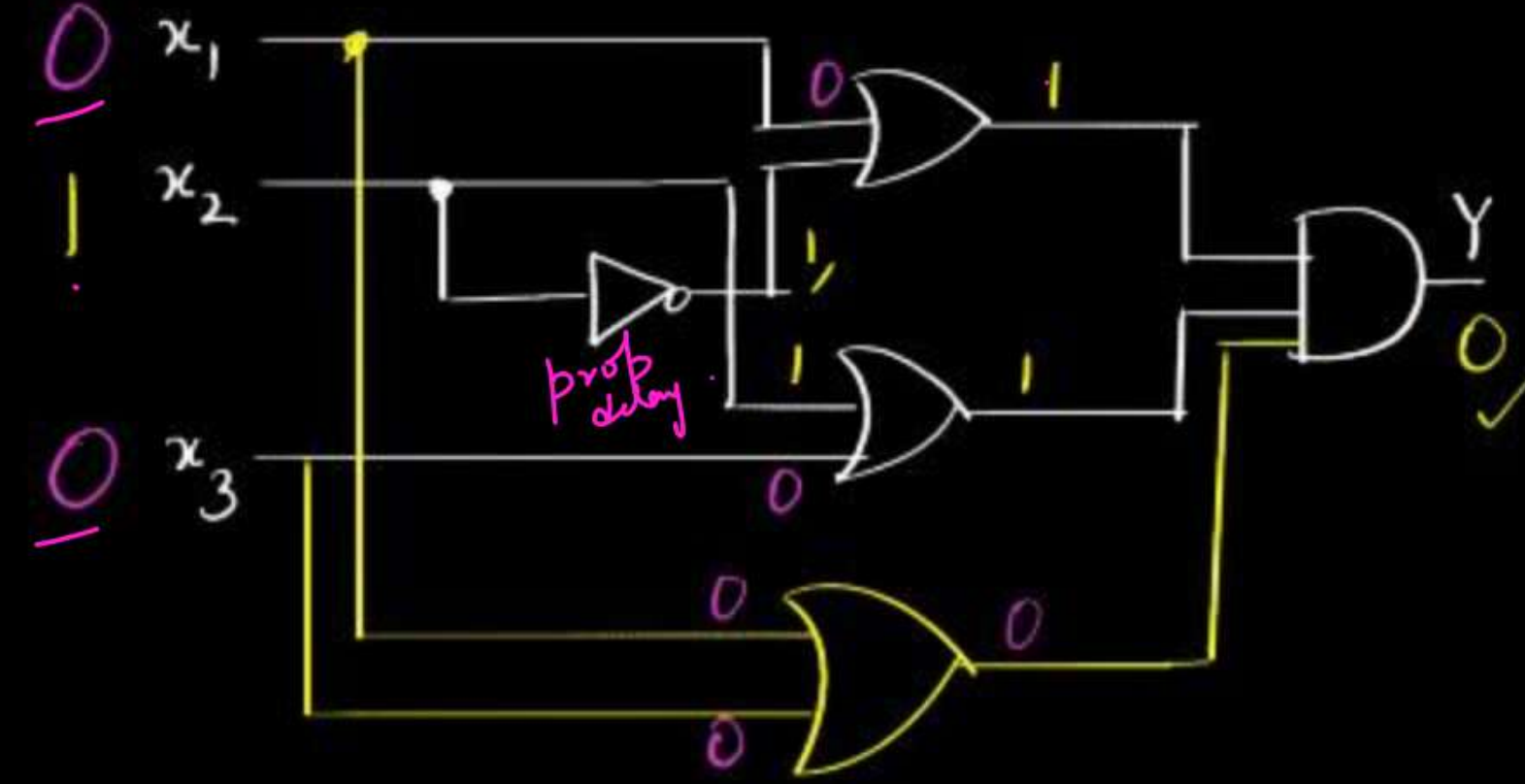
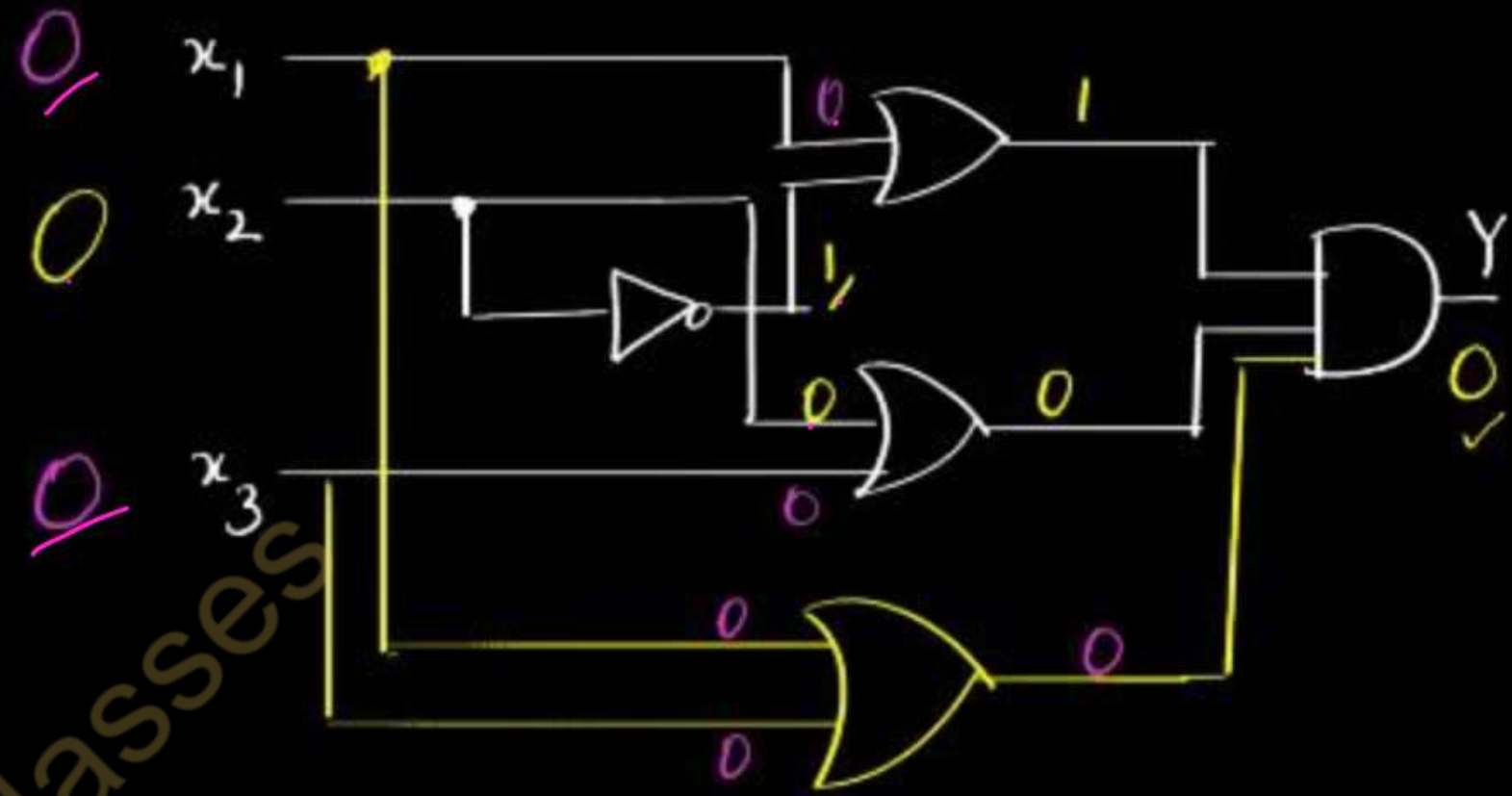
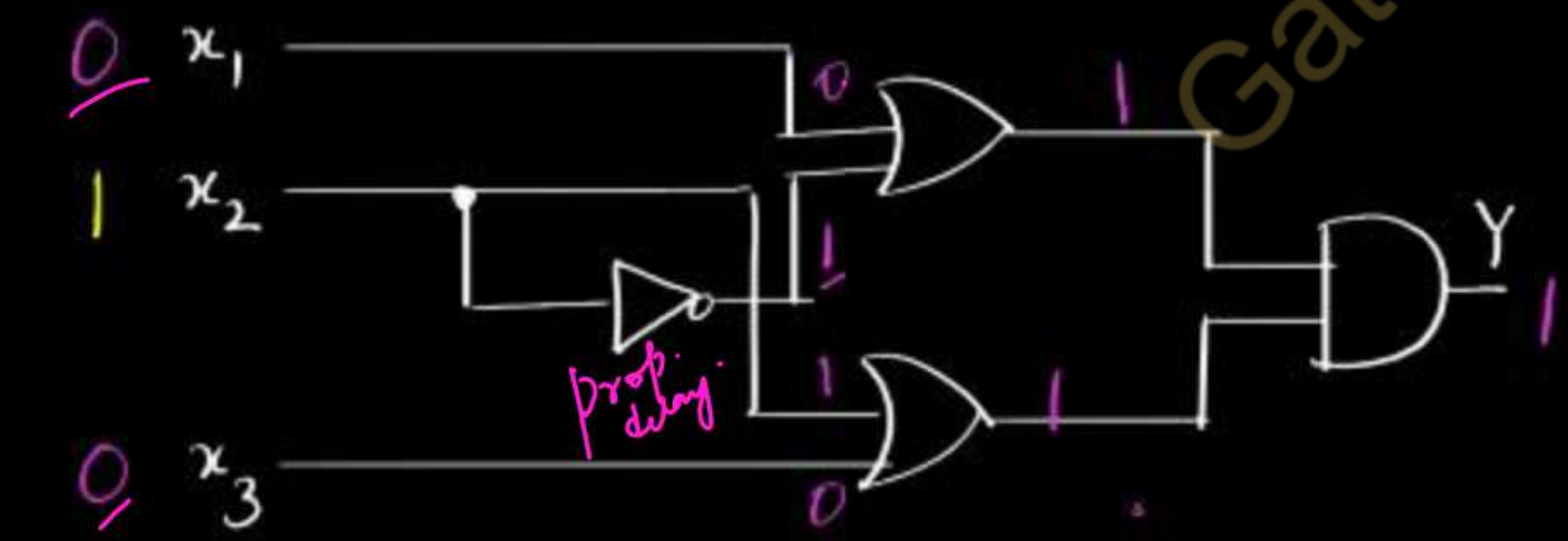
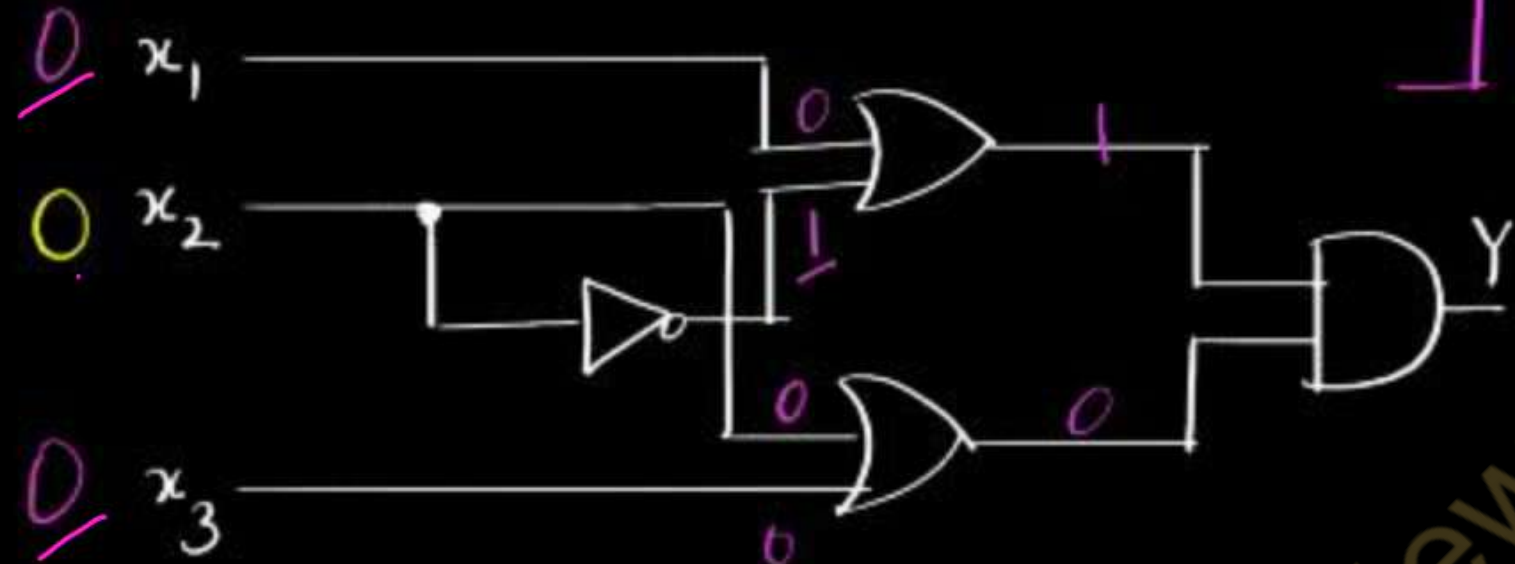
	x_1, x_2	00	01	11	10
0		0	0		0
1			0		

Annotations: $(x_1 + x_3)$ points to the 01 column. $(x_1 + \bar{x}_2)$ points to the 11 column. $(x_2 + x_3)$ points to the 01 column and the 10 column.

$x_3 \backslash x_1 x_2$	00	01	11	10
0	0	0		0
1		0		

$(x_1 + x_3)$ points to the 0 in row 0, column 00.
 $(x_1 + \bar{x}_2)$ points to the 0 in row 0, column 01.
 $(x_2 + x_3)$ points to the 0 in row 1, column 01.

$x_1 = x_3 = 0$





Gateway Classes



Empowering Learners, Transforming Futures

Thank You



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